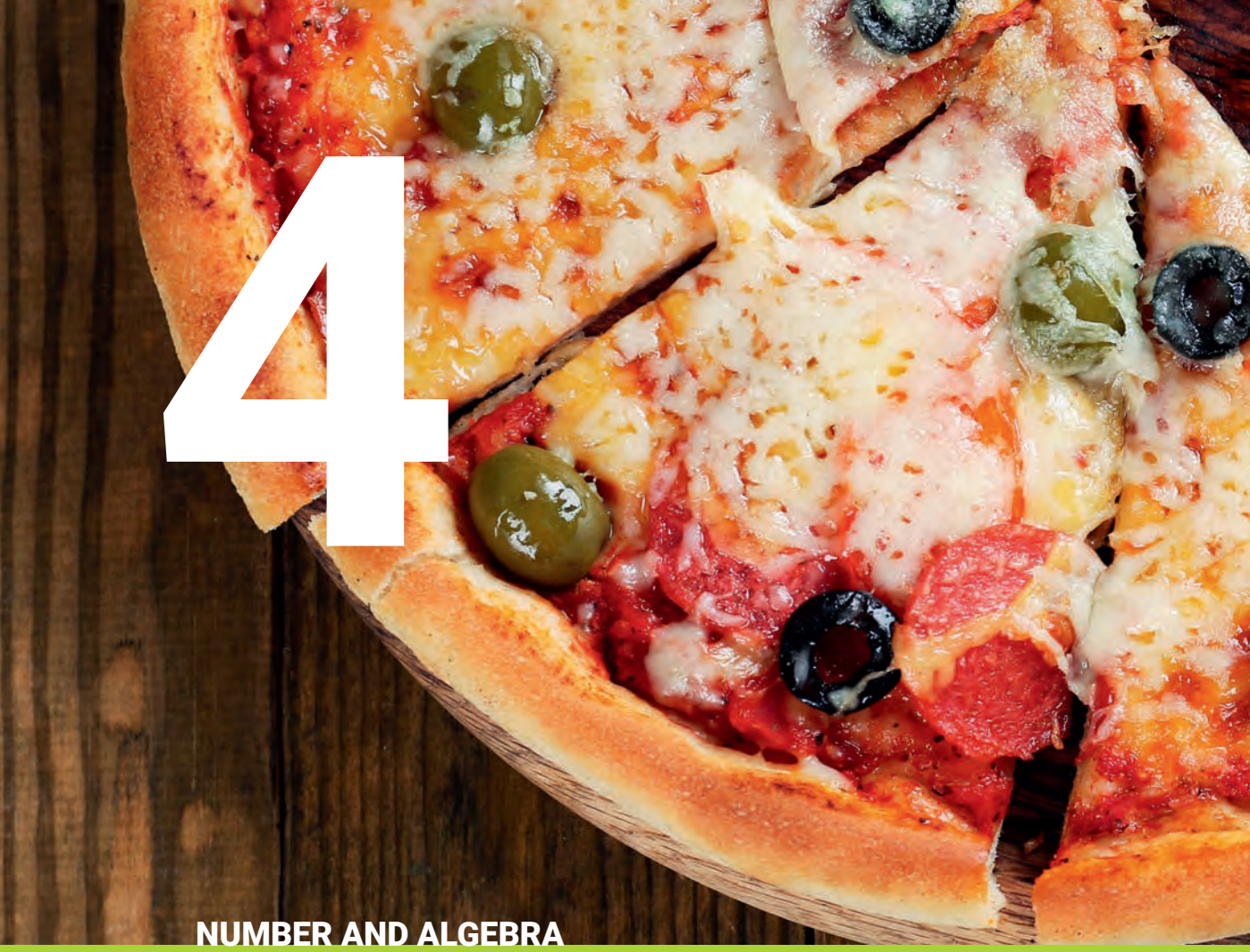


4

NUMBER AND ALGEBRA

FRACTIONS AND PERCENTAGES

We use fractions whenever we want to find part of a whole: when we share pizza or cake, when we measure ingredients in cooking or when we say "it's quarter past 10" to tell the time. The word **fraction** comes from the Latin *fractio*, meaning 'to break'. Chemists use fractional distillation to break chemical compounds in liquid form into their component parts.



Chapter outline

	Working mathematically				
4.01 Fractions	U	F		R	C
4.02 Ordering fractions	U			R	C
4.03 Adding and subtracting fractions	U	F		R	
4.04 Adding and subtracting mixed numerals	U	F	PS	R	
4.05 Fraction of a quantity	U	F	PS	R	C
4.06 Multiplying fractions	U	F	PS	R	C
4.07 Dividing fractions	U	F	PS	R	C
4.08 Fraction problems	U	F	PS	R	
4.09 Fractions and the calculator	U	F	PS	R	
4.10 Percentages, fractions and decimals	U	F			C
4.11 Percentage of a quantity	U	F			
4.12 Expressing quantities as fractions and percentages	U	F			

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Wordbank

denominator The number at the bottom in a fraction

improper fraction A fraction such as $\frac{7}{3}$, in which the numerator is larger than or equal to the denominator

mixed numeral A number such as $3\frac{2}{5}$, made up of a whole number and a fraction

numerator The number at the top in a fraction

percentage A special type of fraction in which the denominator is 100

proper fraction A fraction such as $\frac{4}{10}$ in which the numerator is smaller than the denominator

reciprocal (of a fraction) The fraction 'turned upside down', for example, the reciprocal of $\frac{3}{8}$ is $\frac{8}{3}$

simplify (a fraction) To reduce the numerator and the denominator of a fraction to their lowest form by dividing by the same factor

Quiz
Wordbank 4

Videos (12):

SkillCheck Using fractions

4.01 Improper fractions and mixed numerals • Equivalent fractions • Simplifying fractions

4.03 Working out fractions • Adding and subtracting fractions

4.05 Fraction of a quantity

4.06 Multiplying fractions • Multiplying and dividing fractions

4.07 Dividing fractions • Multiplying and dividing fractions

4.10 Converting percentages to fractions and decimals • Converting fractions and decimals to percentages

4.12 Expressing quantities as fractions and percentages

Twig video (1):

4.05 Fractions: Slow motion

PhET interactives (5):

4.01 Fractions: Intro • Fraction matcher • Build a fraction • Fractions: Mixed numbers

4.02 Fractions: Equality

Quizzes (6):

- Wordbank 4
- SkillCheck 4
- Mental skills 4A
- Mental skills 4B
- Language of maths 4
- Test yourself 4

Skillsheets (4):

SkillCheck Fractions

4.01 Fractions • Improper fractions and mixed numerals • Equivalent fractions • Simplifying fractions

Worksheets (18):

SkillCheck Fraction diagrams

4.01 Classifying fractions 1 • Classifying fractions 2 • Fractions wall • Pop stick calculator • Fraction strips

4.02 Fractions wall • Pop stick calculator

4.03 Pop stick calculator • Magic squares

4.06 Multiplying fractions using rectangles

4.07 What's the fraction question? • Multiplying and dividing fractions number grid

4.10 Percentages wall • Percentage shapes • Fractions, decimals and percentages

4.11 Percentages of an amount

4.12 Calculating percentages • Money and percentages review

Mind map Fractions and percentages

Chapter review Fractions and decimals review

Puzzles (13):

4.01 Mixed numbers to improper fractions • Equivalent fractions • Equivalent fractions 1

4.03 Adding and subtracting fractions

4.05 Fractions of words

4.09 Fraction arithmagons

4.10 Converting percentages to fractions • Converting fractions to percentages • Converting decimals to percentages

4.11 Percentages • Percentages cross number • Percentages of an amount

4.12 Calculating percentages

Chapter review Fractions and percentages crossword

Presentations (3):

4.01 Improper fractions

4.03 Adding and subtracting fractions

4.10 Percentages, fractions and decimals

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

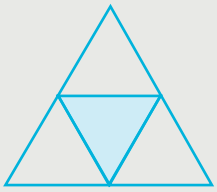
- compare and order fractions, including mixed numerals and negative fractions
- solve problems involving adding, subtracting, multiplying and dividing fractions
- find a fraction of a quantity
- convert between fractions, decimals and percentages
- find a percentage of a quantity
- express one quantity as a fraction or percentage of another

SkillCheck ANSWERS ON P. 571

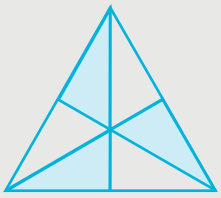
1 For each diagram, state what fraction of the whole:

i is shaded **ii** is unshaded

a



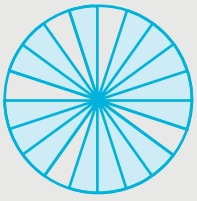
b



c



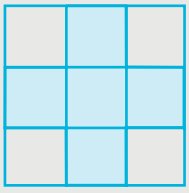
d



e



f



2 What fraction of the month shown on this calendar has been crossed out?

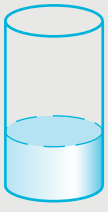
March 2023

Sun	Mon	Tue	Wed	Thu	Fri	Sat
			~~1~~	~~2~~	~~3~~	~~4~~
~~5~~	~~6~~	~~7~~	~~8~~	~~9~~	~~10~~	~~11~~
~~12~~	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

3 Draw a square and divide it into quarters.

4 Draw a rectangle and divide it into thirds.

5 Approximately how full is this glass?



Quiz
SkillCheck 4

Video
Using fractions

Worksheet
Fraction diagrams

Skillsheet
Fractions

6 Find:

a $\frac{1}{2}$ of 24 **b** $\frac{1}{2}$ of 38 **c** $\frac{1}{3}$ of 21

d $\frac{1}{10}$ of 70 **e** $\frac{1}{4}$ of 20 **f** $\frac{1}{5}$ of 15

7 Draw a circle and divide it into eighths.

8 Find the highest common factor (HCF) of each pair of numbers.

a 15 and 18 **b** 9 and 21 **c** 16 and 40

9 Find the lowest common multiple (LCM) of each pair of numbers.

a 3 and 4 **b** 5 and 10 **c** 8 and 6

10 Draw any shape and divide it into fifths.

11 What fraction of one hour is:

a 13 minutes? **b** 45 minutes? **c** 10 minutes?

12 Which fraction is larger: $\frac{1}{3}$ or $\frac{1}{4}$?

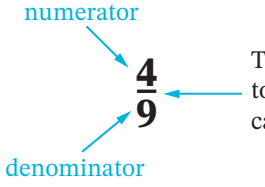
4.01 Fractions

Skillsheet
Fractions

Interactives
Fractions: Intro
Fraction matcher
Build a fraction
Fractions: Mixed numbers

A **fraction** is called a **rational number** because it can be written as $\frac{a}{b}$, where a and b are integers (but b cannot be zero). Its more correct name is **common fraction** because decimals and percentages are also types of fractions.

- The top number (a) is the **numerator**
- The bottom number (b) is the **denominator**



The line between the top and bottom is called the **vinculum**.

Types of fractions

A **proper fraction** has a numerator that is less than the denominator, for example, $\frac{4}{9}, \frac{11}{13}, \frac{37}{60}, \frac{80}{100}$.

An **improper fraction** has a numerator that is more than or equal to the denominator, for example, $\frac{9}{4}, \frac{17}{3}, \frac{15}{15}, \frac{12}{10}$.

A **mixed numeral** is made up of a whole number and a proper fraction, for example, $4\frac{1}{2}, 5\frac{2}{3}, 3\frac{1}{7}, 4\frac{3}{10}$.

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Improper fractions and mixed numerals

An improper fraction is a 'top-heavy' fraction so its value is greater than 1, and it can be converted to a mixed numeral or whole number.

Converting improper fractions to mixed numerals

To convert an improper fraction to a mixed numeral, divide the numerator by the denominator and write the remainder as a proper fraction.

Example 1

Convert $\frac{38}{3}$ to a mixed numeral.

Solution

$\frac{38}{3} = 38 \div 3$

$= 12$ remainder 2

$= 12\frac{2}{3}$

To find how many 'wholes' in 38 thirds, divide 38 by 3.

Write the remainder in the numerator of a fraction.

Example 2

Convert $5\frac{3}{7}$ to an improper fraction.

Solution

$5\frac{3}{7} = 5 + \frac{3}{7}$

$= \frac{5 \times 7}{7} + \frac{3}{7}$

$= \frac{35}{7} + \frac{3}{7}$

$= \frac{38}{7}$

Converting mixed numerals to improper fractions

To convert a mixed numeral to an improper fraction, multiply the denominator by the whole number, then add the numerator terms and write the total as the new numerator of the fraction.

$$5\frac{3}{7} = \frac{7 \times 5 + 3}{7} = \frac{38}{7}$$

Skillsheet Improper fractions and mixed numerals

Worksheets Classifying fractions 1

Classifying fractions 2

Video Improper fractions and mixed numerals

4.01

Video Improper fractions and mixed numerals

Presentation Improper fractions

Puzzle Mixed numbers to improper fractions

Equivalent fractions

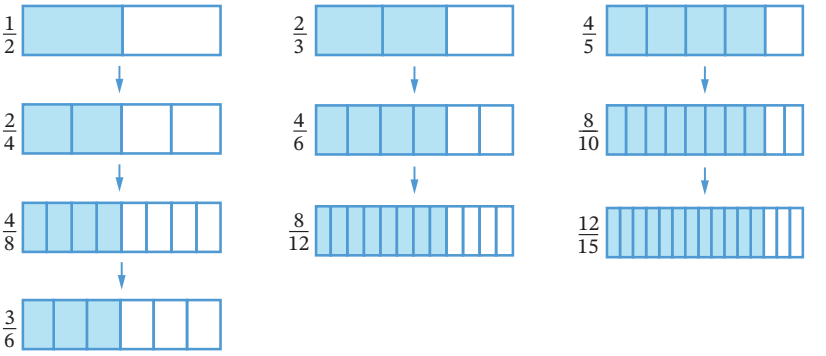
Skillsheet
Equivalent fractions

Worksheets
Fractions wall
Pop stick calculator
Fraction strips

Puzzle
Equivalent fractions

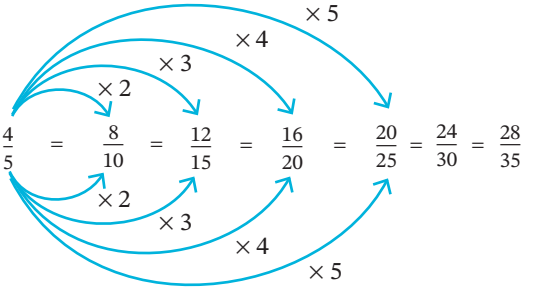
Video
Equivalent fractions

Equivalent fractions are equal. They may have different names, but are worth the same amount. Look at the set of fraction strips below. In each set, the fractions are equivalent.



$\frac{1}{2} = \frac{2}{4} = \frac{4}{8} = \frac{3}{6}$ $\quad \frac{2}{3} = \frac{4}{6} = \frac{8}{12}$ $\quad \frac{4}{5} = \frac{8}{10} = \frac{12}{15}$

This diagram shows fractions that are **equivalent** (equal) to $\frac{4}{5}$.



Equivalent fractions

Equivalent fractions are found by multiplying or dividing the numerator and the denominator by the same number.

Example 3

For each pair of equivalent fractions, find the missing number.

a $\frac{3}{4} = \frac{?}{20}$ **b** $\frac{5}{9} = \frac{35}{?}$ **c** $\frac{16}{24} = \frac{?}{6}$

Solution

a To find the missing numerator, look at the 2 denominators, 4 and 20.

$\frac{3}{4} = \frac{?}{20}$

4 is multiplied by 5 to give 20, so do the same thing to the numerator 3.

$\frac{3}{4} = \frac{3 \times 5}{4 \times 5} = \frac{15}{20}$

The missing number is 15.

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b To find the missing denominator, look at the numerators 5 and 35.

$\frac{5}{9} = \frac{35}{?}$

5 is multiplied by 7 to give 35, so do the same to the 9.

$\frac{5}{9} = \frac{5 \times 7}{9 \times 7} = \frac{35}{63}$

The missing number is 63.

c To find the missing numerator, look at the denominators 24 and 6.

$\frac{16}{24} = \frac{?}{6}$

24 is **divided** by 4 to give 6, so do the same to 16.

$\frac{16}{24} = \frac{16 \div 4}{24 \div 4} = \frac{4}{6}$

The missing number is 4.

Simplifying fractions

We have seen the equivalent fractions $\frac{4}{5} = \frac{8}{10} = \frac{12}{15}$ described above. The most basic fraction, $\frac{4}{5}$, is called the **simplified fraction** or the fraction **reduced** to its **lowest form**. A simplified fraction has the smallest numerator and denominator possible.

Simplifying fractions

- To simplify a fraction, keep dividing the numerator and denominator by the same number (preferably a large number such as their highest common factor (HCF))
- If the numerator and denominator are both even, divide by 2 or perhaps 4
- Otherwise, try dividing by the odd numbers 3, 5 or 7

Example 4

Simplify each fraction.

a $\frac{20}{30}$ **b** $\frac{16}{48}$ **c** $\frac{36}{27}$

Solution

a $\frac{20}{30} = \frac{20 \div 10}{30 \div 10} = \frac{2}{3}$

b $\frac{16}{48} = \frac{16 \div 2}{48 \div 2} = \frac{8}{24} = \frac{8 \div 8}{24 \div 8} = \frac{1}{3}$

or $\frac{16}{48} = \frac{16 \div 16}{48 \div 16} = \frac{1}{3}$

c $\frac{36}{27} = \frac{36 \div 9}{27 \div 9} = \frac{4}{3} = 1\frac{1}{3}$

4.01

Skillsheet Simplifying fractions

Worksheet Pop stick calculator

Puzzle Equivalent fractions 1

Video Simplifying fractions

EXERCISE 4.01 ANSWERS ON P. 571

Fractions U F R C

1 Classify each fraction as being proper (P), improper (I) or a mixed numeral (M). C

a $1\frac{1}{2}$ **b** $\frac{3}{100}$ **c** $3\frac{2}{5}$ **d** $20\frac{1}{4}$

e $\frac{2}{3}$ **f** $\frac{100}{45}$ **g** $\frac{150}{250}$ **h** $\frac{86}{87}$

i $\frac{3}{2}$ **j** $\frac{3}{5}$ **k** $4\frac{1}{20}$ **l** $\frac{1000}{25}$

EXAMPLE 1

2 Convert each improper fraction to a mixed numeral or whole number.

a $\frac{13}{3}$ **b** $\frac{13}{4}$ **c** $\frac{13}{10}$ **d** $\frac{21}{5}$

e $\frac{38}{35}$ **f** $\frac{35}{8}$ **g** $\frac{30}{6}$ **h** $\frac{36}{5}$

i $\frac{28}{7}$ **j** $\frac{120}{11}$ **k** $\frac{72}{19}$ **l** $\frac{101}{10}$

EXAMPLE 2

3 Convert each mixed numeral to an improper fraction.

a $1\frac{1}{2}$ **b** $1\frac{5}{8}$ **c** $1\frac{2}{7}$ **d** $3\frac{1}{3}$

e $4\frac{1}{2}$ **f** $3\frac{7}{10}$ **g** $6\frac{1}{6}$ **h** $7\frac{3}{5}$

EXAMPLE 3

4 Complete each set of equivalent fractions.

a $\frac{1}{4}=\frac{3}{?}$ **b** $\frac{5}{6}=\frac{?}{30}$ **c** $\frac{2}{3}=\frac{?}{12}$

d $\frac{7}{10}=\frac{?}{80}$ **e** $\frac{7}{8}=\frac{63}{?}$ **f** $\frac{3}{4}=\frac{?}{100}$

g $\frac{20}{24}=\frac{5}{?}$ **h** $\frac{15}{40}=\frac{?}{8}$ **i** $\frac{18}{32}=\frac{?}{16}$

5 Write 3 equivalent fractions for each fraction.

a $\frac{3}{5}$ **b** $\frac{2}{3}$ **c** $\frac{1}{8}$

6 What is the missing numerator in $\frac{12}{18}=\frac{?}{3}$? Select the correct answer **A**, **B**, **C** or **D**.

A 6 **B** 1 **C** 5 **D** 2

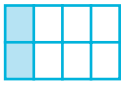
EXAMPLE 4

7 Simplify each fraction.

a $\frac{6}{16}$ **b** $\frac{6}{24}$ **c** $\frac{40}{50}$ **d** $\frac{24}{32}$

e $\frac{18}{42}$ **f** $\frac{60}{75}$ **g** $\frac{5}{20}$ **h** $\frac{14}{21}$

8 What fraction of each shape is shaded, in simplest form? Select **A**, **B**, **C** or **D**.

a 

A $\frac{2}{8}$ **B** $\frac{6}{8}$ **C** $\frac{1}{4}$ **D** $\frac{1}{8}$

b 

A $\frac{1}{3}$ **B** $\frac{4}{6}$ **C** $\frac{3}{2}$ **D** $\frac{2}{3}$

Foundation Standard Complex

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c 

A $\frac{4}{8}$ **B** $\frac{2}{4}$ **C** $\frac{1}{2}$ **D** $\frac{8}{16}$

9 Convert each improper fraction to a mixed numeral in simplest form.

a $\frac{18}{8}$ **b** $\frac{30}{14}$ **c** $\frac{50}{25}$ **d** $\frac{90}{54}$

e $\frac{28}{12}$ **f** $\frac{75}{60}$ **g** $\frac{32}{24}$ **h** $\frac{48}{36}$

10 Write each mixed numeral in simplest form.

a $1\frac{4}{8}$ **b** $3\frac{5}{30}$ **c** $4\frac{5}{20}$ **d** $6\frac{30}{40}$

e $17\frac{35}{100}$ **f** $100\frac{44}{64}$ **g** $1\frac{54}{100}$ **h** $5\frac{4}{12}$

11 Which fraction is not equivalent to $\frac{3}{4}$? Select **A**, **B**, **C** or **D**.

A $\frac{12}{16}$ **B** $\frac{6}{8}$ **C** $\frac{24}{32}$ **D** $\frac{10}{18}$

12 Which is larger: a proper fraction or a mixed numeral? Explain your answer. R C

Ordering fractions 4.02

- Is $\frac{4}{10}$ larger than $\frac{3}{8}$?
- Is $1\frac{2}{3}$ smaller than $1\frac{5}{9}$?

The best way to **compare** fractions with different denominators is to convert them to equivalent fractions with the same denominator first. A common denominator can be found by multiplying the denominators of both fractions together or by using the lowest common multiple (LCM) of the denominators, also called the **lowest common denominator** (LCD).

Worksheets
Fractions wall
Pop stick calculator

Interactive
Fractions: Equality

Example 5

a Which fraction is larger, $\frac{4}{10}$ or $\frac{3}{8}$?

b Which fraction is smaller, $1\frac{2}{3}$ or $1\frac{5}{9}$?

Solution

a **Method 1**

Convert both fractions so that they share a common denominator of $10 \times 8 = 80$.

$\frac{4}{10} = \frac{4 \times 8}{10 \times 8} = \frac{32}{80}$ and $\frac{3}{8} = \frac{3 \times 10}{8 \times 10} = \frac{30}{80}$

Note that we multiply the numerator and denominator of each fraction by the denominator of the **other** fraction.

By comparing numerators, $\frac{32}{80} > \frac{30}{80}$.

$\therefore \frac{4}{10}$ is larger.

Foundation Standard Complex

Method 2

Another suitable denominator is the lowest common denominator (LCD) of both denominators.

The LCD of 10 and 8 is 40.

Convert both fractions.

$\frac{4}{10} = \frac{4 \times 4}{10 \times 4} = \frac{16}{40}$ and $\frac{3}{8} = \frac{3 \times 5}{8 \times 5} = \frac{15}{40}$

By comparing numerators, $\frac{16}{40} > \frac{15}{40}$.

$\therefore \frac{4}{10}$ is larger.

b Since both whole numbers are 1, we only compare the fraction parts.

Method 1

Common denominator $= 3 \times 9 = 27$.

Convert both fractions so that they share a common denominator.

$\frac{2}{3} = \frac{2 \times 9}{3 \times 9} = \frac{18}{27}$ and $\frac{5}{9} = \frac{5 \times 3}{9 \times 3} = \frac{15}{27}$

By comparing numerators, $\frac{15}{27} < \frac{18}{27}$.

$\therefore 1\frac{5}{9}$ is smaller.

Method 2

We only need to convert the first fraction because the second fraction already has 9 as a denominator.

$\frac{2}{3} = \frac{2 \times 3}{3 \times 3} = \frac{6}{9}$ and $\frac{5}{9}$

By comparing numerators, $\frac{5}{9} < \frac{6}{9}$.

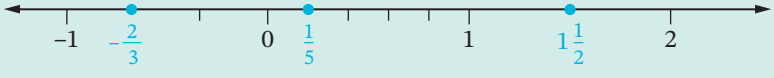
$\therefore 1\frac{5}{9}$ is smaller.

Example 6

Draw a number line and mark in the fractions $\frac{1}{5}$, $-\frac{2}{3}$ and $1\frac{1}{2}$.

Solution

- Divide the interval between 0 and 1 into fifths and mark $\frac{1}{5}$
- Divide the interval between 0 and –1 into thirds and mark $-\frac{2}{3}$
- Divide the interval between 1 and 2 into halves and mark $1\frac{1}{2}$



EXERCISE 4.02 ANSWERS ON P. 571

Ordering fractions UFRC

EXAMPLE 5

1 For each pair of fractions, find the larger fraction.

a $\frac{7}{10}, \frac{3}{10}$ **b** $\frac{5}{3}, \frac{2}{3}$ **c** $\frac{3}{4}, \frac{7}{8}$

d $\frac{2}{3}, \frac{3}{10}$ **e** $2\frac{1}{5}, 2\frac{1}{6}$ **f** $3\frac{4}{7}, 2\frac{5}{6}$

2 For each pair of fractions, find the smaller fraction.

a $\frac{4}{5}, \frac{3}{5}$ **b** $\frac{11}{8}, \frac{9}{8}$ **c** $\frac{1}{4}, \frac{3}{10}$

d $\frac{1}{6}, \frac{2}{3}$ **e** $4\frac{3}{4}, 4\frac{2}{3}$ **f** $1\frac{2}{7}, 1\frac{1}{5}$

3 Which set of fractions shows $\frac{19}{16}, \frac{7}{6}, \frac{5}{4}, \frac{33}{32}$ in ascending order?
Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{7}{6}, \frac{5}{4}, \frac{19}{16}, \frac{33}{32}$ **B** $\frac{5}{4}, \frac{7}{6}, \frac{19}{16}, \frac{33}{32}$ **C** $\frac{33}{32}, \frac{7}{6}, \frac{19}{16}, \frac{5}{4}$ **D** $\frac{19}{16}, \frac{33}{32}, \frac{7}{6}, \frac{5}{4}$

4 Copy and complete each statement with a $<$ or $>$ symbol. C

a $\frac{1}{2} ____ \frac{1}{3}$ **b** $\frac{2}{3} ____ \frac{3}{4}$ **c** $\frac{3}{8} ____ \frac{1}{2}$

d $2\frac{3}{5} ____ 2\frac{7}{10}$ **e** $\frac{5}{6} ____ \frac{1}{2}$ **f** $\frac{5}{12} ____ \frac{1}{3}$

g $\frac{17}{20} ____ \frac{6}{10}$ **h** $\frac{11}{15} ____ \frac{2}{3}$ **i** $2\frac{3}{7} ____ 5\frac{2}{5}$

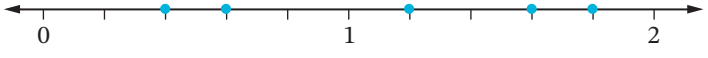
5 Find a proper fraction that is greater than $\frac{3}{4}$:

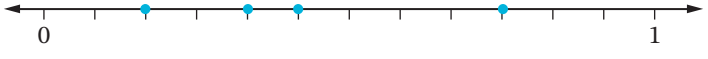
a with a denominator of 10

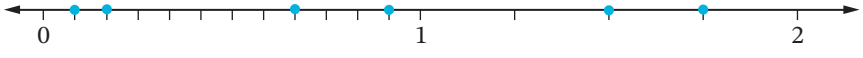
b with any other denominator

6 For each number line, find the value of each fraction marked by a dot.

a


b


c


d


e


EXAMPLE 6

7 Plot each set of fractions on a separate number line.

a $\frac{2}{3}$ $\frac{3}{3}$ $1\frac{1}{3}$ $\frac{7}{3}$ **b** $\frac{1}{5}$ $1\frac{3}{5}$ $\frac{5}{5}$ $2\frac{2}{5}$ $\frac{3}{5}$ $\frac{9}{5}$

c $\frac{5}{6}$ $-\frac{1}{3}$ $\frac{1}{2}$ $-1\frac{1}{6}$ **d** $-\frac{7}{10}$ $\frac{2}{5}$ $-\frac{1}{2}$ $-\frac{1}{4}$ $\frac{9}{10}$

8 Which one of these fractions is closest to $\frac{3}{4}$? Select **A**, **B** or **C**. R

A $\frac{1}{2}$ **B** $\frac{7}{8}$ **C** $\frac{13}{16}$

9 Find a fraction that is between: R

a $\frac{3}{4}$ and $\frac{4}{5}$ **b** $\frac{5}{8}$ and $\frac{2}{3}$

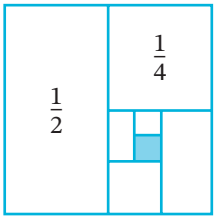
Foundation Standard Complex

Investigation

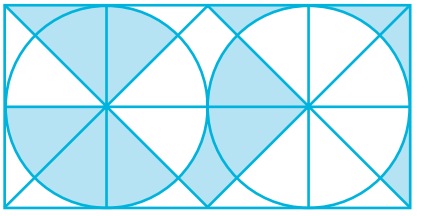
What fraction is shaded?

In groups or individually, decide which fraction of each diagram has been shaded. Explain how you solved each puzzle.

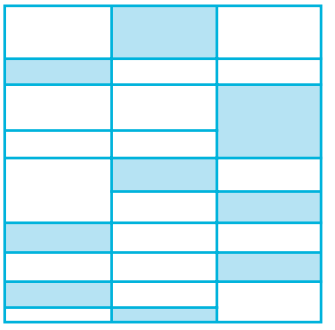
1



2



3



4.03 Adding and subtracting fractions

Video
Working out fractions

Worksheets
Pop stick calculator
Magic squares

Puzzle
Adding and subtracting fractions

Adding and subtracting fractions with the same denominator

To add or subtract fractions with the same denominator, simply add or subtract the numerators.

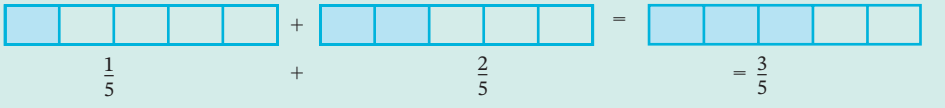
Example 7

Find:

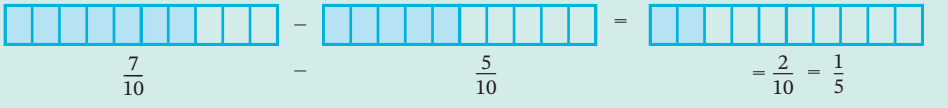
a $\frac{1}{5}+\frac{2}{5}$ **b** $\frac{7}{10}-\frac{5}{10}$

Solution

a



b



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Fractions with different denominators

Adding and subtracting fractions with different denominator

To add or subtract fractions with different denominators, we must first convert them to equivalent fractions with the same denominator.

Example 8

Find:

a $\frac{1}{4}+\frac{5}{6}$ **b** $\frac{2}{5}-\frac{3}{20}$

Solution

a **Method 1**

Common denominator $= 6 \times 4 = 24$.

$\frac{1}{4}=\frac{1\times 6}{4\times 6}=\frac{6}{24}$ and $\frac{5}{6}=\frac{5\times 4}{6\times 4}=\frac{20}{24}$

$$\frac{1}{4}+\frac{5}{6}=\frac{6}{24}+\frac{20}{24}$$
$$=\frac{26\div 2}{24\div 2}$$
$$=\frac{13}{12}$$
$$=1\frac{1}{12}$$

Method 2

The LCD of 4 and 6 is 12.

$\frac{1}{4}=\frac{1\times 3}{4\times 3}=\frac{3}{12}$ and $\frac{5}{6}=\frac{5\times 2}{6\times 2}=\frac{10}{12}$

$$\frac{1}{4}+\frac{5}{6}=\frac{3}{12}+\frac{10}{12}$$
$$=\frac{13}{12}$$
$$=1\frac{1}{12}$$

b **Method 1**

Common denominator $= 5 \times 20 = 100$

$\frac{2}{5}=\frac{2\times 20}{5\times 20}=\frac{40}{100}$ and $\frac{3}{20}=\frac{3\times 5}{20\times 5}=\frac{15}{100}$

$$\frac{2}{5}-\frac{3}{20}=\frac{40}{100}-\frac{15}{100}$$
$$=\frac{25\div 25}{100\div 25}$$
$$=\frac{1}{4}$$

Method 2

The LCD of 5 and 20 is 20.

$\frac{2}{5}=\frac{2\times 4}{5\times 4}=\frac{8}{20}$ and $\frac{3}{20}$

$$\frac{2}{5}-\frac{3}{20}=\frac{8}{20}-\frac{3}{20}$$
$$=\frac{5\div 5}{20\div 5}$$
$$=\frac{1}{4}$$

Video: Adding and subtracting fractions

Presentation: Adding and subtracting fractions

4.03

EXERCISE 4.03 ANSWERS ON P. 572

Adding and subtracting fractions U F R

EXAMPLE 7

1 Evaluate each expression.

a $\frac{3}{6}+\frac{1}{6}$ **b** $\frac{7}{10}+\frac{2}{10}$ **c** $\frac{1}{2}+\frac{1}{2}$ **d** $\frac{2}{3}+\frac{2}{3}$

e $\frac{3}{5}-\frac{1}{5}$ **f** $\frac{5}{4}-\frac{3}{4}$ **g** $\frac{9}{10}-\frac{5}{10}$ **h** $\frac{7}{12}-\frac{4}{12}$

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EXAMPLE 8

2 Evaluate each expression.

a $\frac{1}{2}+\frac{1}{3}$ **b** $\frac{2}{3}+\frac{1}{5}$ **c** $\frac{3}{4}-\frac{1}{2}$ **d** $\frac{5}{8}-\frac{1}{2}$

e $\frac{2}{6}+\frac{1}{3}$ **f** $\frac{4}{5}-\frac{3}{4}$ **g** $\frac{3}{7}-\frac{1}{9}$ **h** $\frac{7}{12}+\frac{1}{6}$

i $\frac{5}{8}+\frac{1}{4}$ **j** $\frac{9}{10}-\frac{3}{5}$ **k** $\frac{7}{9}-\frac{2}{3}$ **l** $\frac{5}{6}+\frac{3}{8}$

m $\frac{7}{8}-\frac{2}{10}$ **n** $\frac{3}{4}+\frac{4}{5}$ **o** $\frac{5}{8}-\frac{1}{3}$ **p** $\frac{6}{7}+\frac{1}{5}$

3 What fraction goes in the blank? $\frac{7}{8}-$ ____ $=\frac{1}{4}$. Select the correct answer **A**, **B**, **C** or **D**. R

A $\frac{6}{4}$ **B** $\frac{8}{12}$ **C** $\frac{3}{8}$ **D** $\frac{5}{8}$

4 Copy and complete each equation with the correct fraction. R

a $\frac{1}{2}+$ ____ $=\frac{2}{3}$ **b** ____ $-\frac{1}{4}=\frac{2}{3}$

5 A bottle is $\frac{2}{3}$ full. $\frac{1}{4}$ of the bottle is then poured out. What fraction remains in the bottle? R

6 Evaluate each expression.

a $4+\frac{1}{2}$ **b** $3\frac{2}{3}-3$ **c** $4\frac{1}{6}-\frac{1}{6}$ **d** $5\frac{3}{4}-\frac{1}{4}$

e $2\frac{3}{5}+\frac{4}{5}$ **f** $6\frac{5}{8}-\frac{2}{8}$ **g** $\frac{5}{6}+1\frac{5}{6}$ **h** $3\frac{4}{5}+2\frac{1}{5}$

7 A cash prize is shared between 3 girls. Cindy receives $\frac{1}{4}$, Carmen gets $\frac{2}{5}$ and Libby takes the rest. What fraction does Libby take? R

8 Copy and complete each equation using fractions with different denominators. R

a ____ + ____ $=\frac{15}{16}$ **b** ____ − ____ $=\frac{3}{4}$

Did you know?

Ancient fractions

The earliest mention of fractions comes from ancient Egypt in 1800 BCE. The Egyptians wrote all their fractions as unit fractions, using 1 as their numerator. The Egyptians drew a mouth above a number to show that it was a fraction. For example,  meant $\frac{1}{4}$.

Around 100 CE, the ancient Romans used words to describe fractions. For example:

- $\frac{1}{12}$ was called **uncia** (where the words 'ounce' and 'inch' come from)
- $\frac{6}{12}$ was called **semis**

By 500 CE, Indian mathematicians were writing fractions almost like we do, with a numerator above a denominator, but without a line. The Arabs added the line (vinculum), which we now use to separate the numerator and denominator.

Indian	Arab
5	$=\frac{8}{17}$
−7	

What does 'semi-' mean?

Why do you think the ancient Romans used the word 'semis' for $\frac{6}{12}$?

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Adding and subtracting mixed numerals 4.04

Adding and subtracting mixed numerals

To add or subtract mixed numerals, add or subtract the whole numbers first and then add or subtract the fractions.

Example 9

Find:

a $6\frac{1}{3}+4\frac{1}{2}$

b $10\frac{3}{4}-3\frac{2}{5}$

Solution

a
$$\begin{aligned}6\frac{1}{3}+4\frac{1}{2}&=6+4+\frac{1}{3}+\frac{1}{2}\\&=10+\frac{2}{6}+\frac{3}{6}\\&=10\frac{5}{6}\end{aligned}$$

b
$$\begin{aligned}10\frac{3}{4}-3\frac{2}{5}&=10-3+\frac{3}{4}-\frac{2}{5}\\&=7+\frac{15}{20}-\frac{8}{20}\\&=7\frac{7}{20}\end{aligned}$$

Example 10

Find:

a $7-\frac{3}{5}$

b $5\frac{2}{3}-3\frac{3}{4}$

Solution

a
$$\begin{aligned}7-\frac{3}{5}&=6+1-\frac{3}{5}\\&=6+\frac{5}{5}-\frac{3}{5}\\&=6\frac{2}{5}\end{aligned}$$

b
$$\begin{aligned}5\frac{2}{3}-3\frac{3}{4}&=5+\frac{2}{3}-3\frac{3}{4}\\&=5-3+\frac{2}{3}-\frac{3}{4}\\&=2+\frac{8}{12}-\frac{9}{12}\\&=2-\frac{1}{12}\\&=1+\frac{12}{12}-\frac{1}{12}\\&=1\frac{11}{12}\end{aligned}$$

EXERCISE 4.04 ANSWERS ON P. 572

Adding and subtracting mixed numerals U F PS R

EXAMPLE 9

1 Simplify each expression.

a $1\frac{1}{2}+1\frac{1}{2}$ **b** $2\frac{1}{2}+1\frac{3}{4}$ **c** $5\frac{1}{3}-2\frac{1}{5}$ **d** $4\frac{3}{8}-1\frac{1}{4}$

e $1\frac{1}{3}+2\frac{3}{4}$ **f** $2\frac{3}{5}+1\frac{1}{2}$ **g** $4\frac{5}{6}-2\frac{1}{3}$ **h** $1\frac{1}{2}+2\frac{1}{3}$

i $3\frac{3}{10}-2\frac{1}{4}$ **j** $2\frac{3}{4}+1\frac{2}{5}$ **k** $3\frac{1}{3}-1\frac{1}{4}$ **l** $3\frac{1}{7}+2\frac{1}{3}$

2 Evaluate $2\frac{3}{5}+1\frac{1}{3}$. Select the correct answer **A**, **B**, **C** or **D**.

A $3\frac{3}{8}$ **B** $3\frac{2}{5}$ **C** $3\frac{14}{15}$ **D** $3\frac{9}{15}$

EXAMPLE 10

3 Evaluate each difference.

a $3-\frac{1}{2}$ **b** $2-\frac{3}{5}$ **c** $1-\frac{4}{7}$ **d** $4-\frac{2}{3}$

e $2-\frac{1}{4}$ **f** $3-\frac{5}{12}$ **g** $14-\frac{7}{9}$ **h** $22-\frac{17}{20}$

4 **a** $3\frac{1}{5}-1\frac{1}{2}$ **b** $3\frac{1}{4}-2\frac{1}{2}$ **c** $6\frac{2}{5}-4\frac{7}{8}$

d $4\frac{3}{4}-2\frac{4}{5}$ **e** $2\frac{1}{3}-1\frac{1}{2}$ **f** $4\frac{3}{5}-2\frac{9}{10}$

g $3\frac{1}{6}-1\frac{2}{3}$ **h** $5\frac{3}{10}-2\frac{1}{2}$ **i** $5\frac{2}{3}-3\frac{3}{4}$

5 Rojesh drives $\frac{3}{4}$ of an hour to the airport, flies on a plane for $2\frac{1}{2}$ hours, then catches a shuttle bus to the hotel for $\frac{2}{3}$ of an hour. How long does the whole trip take? PS

6 Copy and complete each equation with the correct mixed numeral. R

a $3\frac{1}{2}+____=5\frac{3}{5}$ **b** $4\frac{3}{4}-____=1\frac{1}{8}$

7 Elise participated in a mini-triathlon on the weekend. Her time for the whole event was 6 hours. She swam for $1\frac{3}{4}$ hours and she cycled for $2\frac{2}{5}$ hours. How long did the running section take her? PS R

8 Each Year 7 student at Nelson Secondary College must choose one sport from tennis, volleyball and gymnastics. If $\frac{3}{10}$ of students chose tennis and $\frac{2}{5}$ chose volleyball, what fraction of Year 7 chose gymnastics? R

9 In a magic square, each row, column and diagonal must add to the same total. Copy and complete each magic square with appropriate fractions. R

a

4	$1\frac{1}{2}$	
$\frac{1}{2}$		$4\frac{1}{2}$
3		

b

$\frac{1}{6}$		
	$\frac{2}{3}$	
	$\frac{1}{3}$	$1\frac{1}{6}$

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Fraction of a quantity

4.05

What is $\frac{2}{3}$ of 12? Here are 12 lollies:



If these lollies were divided into 3 equal piles, each pile would be $\frac{1}{3}$ of 12.



$\underbrace{\frac{1}{3}\text{ of } 12 = 4 \quad \frac{1}{3}\text{ of } 12 = 4}_{\frac{2}{3} \times 12 = 8}$

Note: The word 'of' usually means to multiply.

Puzzle
Fractions of words

Video
Fractions: slow motion

Example 11

Find:

a $\frac{1}{4} \times 28$

b $\frac{2}{3}$ of \$30

Solution

a $\frac{1}{4} \times 28 = 28 \div 4$
$= 7$

b $\frac{2}{3}\text{ of } 30 = \left(\frac{1}{3} \times 30\right) \times 2$
$= \$10 \times 2$
$= \$20$

Video
Fraction of a quantity

Example 12

Find:

a $\frac{1}{10}$ of 2 km (in metres)

b $\frac{3}{5}$ of 1 minute

Solution

a $\frac{1}{10}\text{ of } 2\text{ km} = \frac{1}{10} \times 2000\text{ m}$
$= 200\text{ m}$ — Convert 2 km to metres

b $\frac{3}{5}\text{ of } 1\text{ min} = \frac{3}{5}\text{ of } 60\text{ s}$
$= \left(\frac{1}{5} \times 60\text{ s}\right) \times 3$
$= 12\text{ s} \times 3$
$= 36\text{ s}$ — Convert 1 minute to 60 seconds

Video
Fraction of a quantity

EXERCISE 4.05 ANSWERS ON P. 572

Fraction of a quantity U F PS R C

EXAMPLE 11

1 Evaluate each product.

a $\frac{1}{4} \times 36$ **b** $\frac{1}{5} \times 35$ **c** $\frac{2}{5} \times 50$ **d** $\frac{1}{2} \times 16$

e $\frac{5}{12}$ of 12 **f** $\frac{2}{3}$ of 33 **g** $\frac{1}{9}$ of 27 **h** $\frac{6}{9}$ of 45

i $\frac{3}{10} \times 50$ **j** $\frac{5}{6} \times 60$ **k** $\frac{3}{4} \times 100$ **l** $\frac{2}{7} \times 42$

2 Evaluate each product.

a $\frac{1}{2}$ of \$42 **b** $\frac{2}{3}$ of \$12 **c** $\frac{1}{3}$ of \$24 **d** $\frac{1}{3}$ of 60c

e $\frac{1}{4}$ of \$1.20 **f** $\frac{1}{4}$ of \$10 **g** $\frac{3}{8}$ of \$360 **h** $\frac{2}{5}$ of 90c

i $\frac{5}{6}$ of \$3 **j** $\frac{3}{4}$ of \$16 **k** $\frac{4}{5}$ of \$4000 **l** $\frac{3}{10}$ of \$800

EXAMPLE 12

3 Evaluate each amount.

a $\frac{1}{4}$ of 2 hours (in minutes) **b** $\frac{3}{4}$ of 3 hours (in minutes)

c $\frac{3}{4}$ of 1 day (in hours) **d** $\frac{3}{5}$ of half an hour (in minutes)

e $\frac{1}{5}$ of 1 year (in days) **f** $\frac{1}{4}$ of 3 minutes (in seconds)

g $\frac{2}{3}$ of 5 minutes (in seconds) **h** $\frac{3}{10}$ of 1 hour (in minutes)

i $\frac{2}{3}$ of 2 days (in hours) **j** $\frac{1}{3}$ of 2 years (in months)

4 In a squad of 45 swimmers, 3 out of every 5 were female.

a How many are female? **b** How many are male?

5 Evaluate each amount.

a $\frac{1}{4}$ of 1 km (in m) **b** $\frac{3}{4}$ of 3 km (in m) **c** $\frac{3}{4}$ of 6 cm (in mm)

d $\frac{1}{20}$ of 2 m (in cm) **e** $\frac{1}{4}$ of 3 kg (in g) **f** $\frac{7}{10}$ of 5 L (in mL)

g $\frac{1}{5}$ of 1.6 km (in m) **h** $\frac{3}{10}$ of 1 cm (in mm) **i** $\frac{3}{8}$ of 7.2 kg (in g)

j $\frac{5}{8}$ of 1.6 kL (in L) **k** $\frac{3}{5}$ of 8 m (in mm) **l** $\frac{2}{3}$ of 60 t (in kg)

6 Mr Arun began to read a 280-page book. On the weekend, he read $\frac{6}{7}$ of the book.
How many more pages does he have to read before he finishes the book?
Select **A**, **B**, **C** or **D**. PS R

A 230 **B** 46 **C** 240 **D** 40

7 Rana has a pencil box that is $\frac{1}{2}$ full. She puts 4 extra pencils in and the box is now $\frac{2}{3}$ full.
How many pencils will fit into her pencil box? R

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8 Copy and complete each equation. R

a $\frac{3}{4} \times ____ = 9$ **b** $____ \times 24 = 16$

9 Some lollies were placed on a table.
Karin takes away half of the pile.
Gino then takes one-third of what was left.
Heather takes 2 and eats them.
Half of those left are red and there are 8 red lollies. How many lollies were there originally? PS R



Shutterstock.com/Jim Bosiak

10 A farmer died and in his will he left 17 horses to his three children. Half of the horses would go to Ashleigh, the eldest, one-third would go to Bethany, the middle child, and one-ninth would go to Thomas, the youngest. The family was puzzled because you cannot take those fractions of 17 horses without cutting up a horse!

The farmer's wife had an idea. She borrowed a horse from a neighbouring farm to make a total of 18 horses. Now it was possible to divide up the horses according to the farmer's will. PS R C

a How many horses did each child receive?

b Why was the farmer's wife able to return the extra horse?

c What was mathematically incorrect about the farmer's will? Can you work it out?

4.05

Mental skills 4A: Maths without calculators ANSWERS ON P. 572

Quiz
Mental skills 4A

Fraction of a quantity

Learn these commonly-used fractions and their decimal equivalents.

Fraction	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{20}$	$\frac{2}{5}$
Decimal	0.5	0.25	0.125	0.75	0.2	0.1	0.05	0.4

Now we will use them to find a fraction or decimal of a quantity.

1 Study each example.

a $\frac{1}{4} \times 72 = 72 \div 4$
$= 18$

b $\frac{2}{5} \times 40 = \left(\frac{1}{5} \times 40\right) \times 2$
$= 8 \times 2$
$= 16$

c $\frac{3}{4} \times 32 = \left(\frac{1}{4} \times 32\right) \times 3$
$= 8 \times 3$
$= 24$

d $0.5 \times 66 = \frac{1}{2} \times 66$
$= 33$

e $0.05 \times 80 = \frac{1}{20} \times 80$
$= 4$

f $0.125 \times 56 = \frac{1}{8} \times 56$
$= 7$

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2 Now simplify each expression.

a $\frac{1}{2} \times 28$	**b** $\frac{1}{4} \times 36$	**c** $\frac{1}{10} \times 70$	**d** $\frac{1}{8} \times 64$
e $\frac{1}{5} \times 15$	**f** $\frac{1}{10} \times 80$	**g** $\frac{2}{5} \times 25$	**h** $\frac{1}{20} \times 100$
i $\frac{3}{4} \times 44$	**j** $\frac{1}{8} \times 40$	**k** 0.25×60	**l** 0.4×45
m 0.1×260	**n** 0.125×48	**o** 0.75×48	**p** 0.05×120
q 0.2×70	**r** 0.5×320	**s** 0.25×56	**t** 0.125×16

Investigation

Netball scores

Elvira and Christina play netball for the Jets. A netball game is split into quarters and each goal is worth one point. Last Saturday, Elvira and Christina scored all of the Jets' goals for the first and second quarters.

1 In the first quarter, Elvira shot $\frac{1}{4}$ of the goals and Christina shot 12 goals. How many points did the team score in the first quarter?

2 In the second quarter, Christina scored $\frac{1}{5}$ of the team's 20 points while Elvira scored the rest. How many points did Elvira score?

3 What happened in the last 2 quarters? Finish off the story of the netball game by writing 2 more stories involving fractions.



Alamy Stock Photo/The Photolibrary Wales

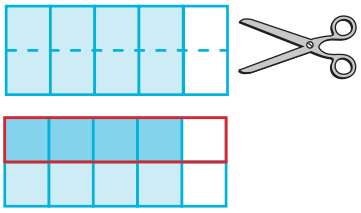
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Multiplying fractions

4.06

What is $\frac{1}{2}$ of $\frac{4}{5}$?

$\frac{4}{5}$ of this diagram is shaded.



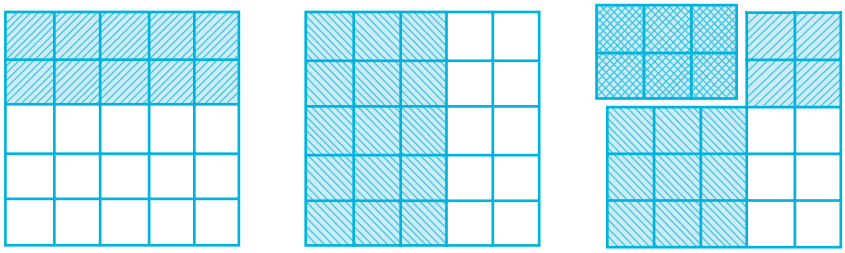
If the diagram were cut in half, each fifth would also be cut in half to make a total of 10 pieces, with 8 of them shaded.

So $\frac{1}{2}$ of $\frac{4}{5}$ (shaded) $= \frac{1}{2}$ of $\frac{8}{10} = \frac{4}{10}$

Another way to work this out is:

$$\frac{1}{2} \text{ of } \frac{4}{5} = \frac{1}{2} \times \frac{4}{5}$$
$$= \frac{1 \times 4}{2 \times 5}$$
$$= \frac{4}{10}$$

You can use diagrams to show other multiplications, for example, $\frac{2}{5} \times \frac{3}{5}$



$\frac{2}{5}$ × $\frac{3}{5}$ = $\frac{6}{25}$

Multiplying fractions

To multiply fractions, multiply the numerators and multiply the denominators separately.

Example 13

Evaluate each product.

a $\frac{3}{5} \times \frac{7}{8}$ **b** $\frac{3}{5} \times \frac{4}{9}$ **c** $\frac{21}{100} \times \frac{8}{15}$

Solution

a $\frac{3}{5} \times \frac{7}{8} = \frac{3 \times 7}{5 \times 8}$

$= \frac{21}{40}$

Worksheet
Multiplying fractions using rectangles

Video
Multiplying and dividing fractions

Video
Multiplying fractions

b $\frac{3}{5} \times \frac{4}{9} = \frac{3 \times 4}{5 \times 9}$

$= \frac{12 \div 3}{45 \div 3}$

$= \frac{4}{15}$

or, by simplifying first before multiplying:

$\frac{3}{5} \times \frac{4}{9} = \frac{\cancel{3}^{1} \times 4}{5 \times \cancel{9}^{3}}$ because $\frac{3}{9} = \frac{1}{3}$

$= \frac{4}{15}$

c By simplifying first before multiplying:

$\frac{21}{100} \times \frac{8}{15} = \frac{\cancel{21}^{7} \times \cancel{8}^{2}}{\cancel{100}^{25} \times \cancel{15}^{5}}$ because $\frac{21}{15} = \frac{7}{5}$, $\frac{8}{100} = \frac{2}{25}$

$= \frac{14}{125}$

Multiplying mixed numerals

Multiplying mixed numerals

To multiply mixed numerals, convert them to improper fractions first.

Example 14

Simplify $3\frac{2}{5} \times 1\frac{2}{3}$.

Solution

$3\frac{2}{5} \times 1\frac{2}{3} = \frac{17}{\cancel{5}^{1}} \times \frac{\cancel{5}^{1}}{3}$

$= \frac{17}{3}$

$= 5\frac{2}{3}$

EXERCISE 4.06 ANSWERS ON P. 572

Multiplying fractions U F PS R C

EXAMPLE 13

1 Evaluate each product.

a $\frac{1}{3} \times \frac{2}{5}$ **b** $\frac{3}{4} \times \frac{3}{5}$ **c** $\frac{5}{6} \times \frac{7}{8}$ **d** $\frac{1}{8} \times \frac{9}{10}$

e $\frac{3}{8} \times \frac{5}{6}$ **f** $\frac{3}{8} \times \frac{3}{8}$ **g** $\frac{4}{9} \times \frac{5}{7}$ **h** $\frac{7}{8} \times \frac{3}{20}$

2 Evaluate $\frac{2}{9} \times \frac{3}{10}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{1}{8}$ **B** $\frac{1}{15}$ **C** $\frac{6}{19}$ **D** $\frac{1}{18}$

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3 Evaluate each product.

a $\frac{1}{5} \times \frac{5}{7}$ **b** $\frac{3}{7} \times \frac{7}{10}$ **c** $\frac{1}{8} \times \frac{4}{9}$ **d** $\frac{3}{7} \times \frac{7}{3}$

e $\frac{3}{8} \times \frac{5}{6}$ **f** $\frac{3}{7} \times \frac{4}{9}$ **g** $\frac{2}{5} \times \frac{3}{8}$ **h** $\frac{3}{8} \times \frac{5}{12}$

i $\frac{3}{10} \times \frac{2}{5}$ **j** $\frac{9}{5} \times \frac{25}{3}$ **k** $\frac{8}{5} \times \frac{10}{7}$ **l** $\frac{1}{6} \times \frac{12}{25}$

m $\frac{5}{6} \times \frac{11}{20}$ **n** $\left(\frac{1}{8}\right)^2$ **o** $\frac{2}{3} \times \frac{9}{10}$ **p** $\left(\frac{3}{4}\right)^2$

4 Convert each mixed numeral to an improper fraction before multiplying.

a $1\frac{2}{3} \times \frac{3}{10}$ **b** $4\frac{1}{3} \times \frac{2}{7}$ **c** $5\frac{2}{5} \times 1\frac{1}{8}$

d $3\frac{5}{6} \times 1\frac{3}{4}$ **e** $1\frac{2}{3} \times 1\frac{2}{3}$ **f** $3\frac{1}{9} \times 1\frac{3}{4}$

5 Mrs Duschevina works $7\frac{3}{4}$ hours a day. She works a full day each day from Monday to Friday and works half a day on Saturday. How many hours a week does she work? PS

6 A cake recipe needs 1 egg, $2\frac{1}{2}$ cups of flour and $\frac{1}{4}$ cup of sugar. What quantities of eggs, flour and sugar will you need to make 8 cakes? R

7 Copy and complete each equation. R C

a $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 3 \times \frac{1}{4}$
$=$

b $\frac{3}{5} + \frac{3}{5} = 2 \times \frac{3}{5}$
$=$

c $\frac{3}{8} + \frac{3}{8} + \frac{3}{8} + \frac{3}{8} + \frac{3}{8} = 5 \times$
$=$
$=$

d $5 \times \frac{1}{7} =$

e $7 \times \frac{3}{20} =$
$=$

f $8 \times \frac{3}{5} =$
$=$

8 Noah wants to be a professional skater. He practises for $2\frac{1}{2}$ hours each day. How many hours of practice does he do in 7 days? PS

9 Copy and complete each equation. R

a $\frac{3}{4} \times ____ = \frac{21}{32}$ **b** $____ \times \frac{1}{5} = \frac{2}{15}$

c $\frac{15}{16} \times ____ = \frac{3}{4}$ **d** $____ \times ____ = \frac{3}{8}$

10 True or false: When a number is multiplied by a fraction, the number is decreased. Give reasons for your answer. R C

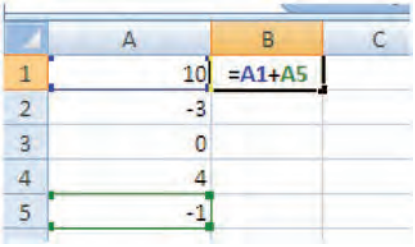
4.06

EXAMPLE 14

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Technology

More spreadsheet practice



	A	B	C
1	10	=A1+A5	
2	-3		
3	0		
4	4		
5	-1		

1 Open a new spreadsheet and enter the 5 numbers shown in column A.

2 In cell B1, enter the formula **=A1+A5** to calculate the sum of the values in cells A1 and A5. You should get the answer 9, because $10 + (-1) = 9$.

3 For the rest of the cells in column B, enter the appropriate formula for each expression shown below and check that the spreadsheet calculates the correct answer. Remember to start each formula with an '=' sign.

a at B2, A2 + A4

b at B3, A2 – A4

c at B4, A3 – A4 – A1

d at B5, A3 – A4 + A1

e at B6, A2 – A5

f at B7, A1 × A2

g at B8, A2 × A4 × A5

h at B9, $(A2)^2$

i at B10, $(A5)^3$

j at B11, A4 ÷ A5

4 At C1, enter a formula for $\frac{A5}{A1}$. When you have obtained the answer, right-click on the cell, select **Format Cells** and convert the answer to a fraction.

5 At C2, enter a formula for $\frac{A1}{A2}$. Right-click on the cell, select **Format Cells** and convert the answer to a mixed numeral.

6 For the rest of the cells in column C, enter the appropriate formula for each expression shown below and check that the spreadsheet calculates the correct answer.

a at C3, the sum of column A

b at C4, the average of column A

c at C5, the minimum value of column A

d at C6, A4 ÷ (–2)

e at C7, A4 ÷ A4

f at C8, (A2 + A4) × A1

g at C9, (A2 – A4) ÷ A5

h at C10, A1 ÷ (A5 – A4)

7 At C11, enter a formula for $\frac{A3}{A2}$. At C12, enter a formula for $\frac{A2}{A3}$. Why doesn't the second formula work?

8 Now enter **new** values in cells A1 to A5 and notice the new answers calculated in columns B and C.

Dividing fractions

4.07

If we turn a fraction upside down and exchange the numerator with the denominator, we get its **reciprocal**.

- The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$ (or $1\frac{1}{2}$)
- The reciprocal of $\frac{1}{4}$ is 4 Note: $\frac{4}{1} = 4$
- The reciprocal of 3 is $\frac{1}{3}$ Note: $3 = \frac{3}{1}$
- The reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$

Note that dividing by 3 is the same as multiplying by $\frac{1}{3}$.

Similarly, dividing by $\frac{1}{3}$ is the same as multiplying by 3.

When dividing by any fraction, we can use its reciprocal.

Worksheets
What's the fraction question?

Multiplying and dividing fractions number grids

Video
Multiplying and dividing fractions

Dividing fractions

To divide by a fraction $\frac{a}{b}$, multiply by its reciprocal $\frac{b}{a}$.

Example 15

Video
Dividing fractions

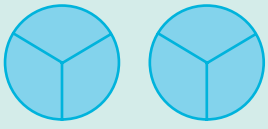
Evaluate each quotient:

a $2 \div \frac{1}{3}$ **b** $\frac{3}{4} \div \frac{1}{2}$ **c** $\frac{7}{3} \div \frac{5}{6}$ **d** $\frac{5}{7} \div 5$

Solution

a $2 \div \frac{1}{3} = 2 \times \frac{3}{1}$
$= 6$

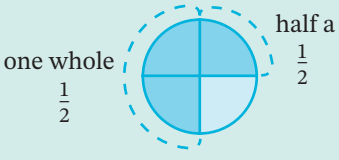
How many times does $\frac{1}{3}$ go into 2?



b $\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times \frac{2}{1}$
$= \frac{6}{4}$
$= \frac{3}{2}$
$= 1\frac{1}{2}$

$\frac{3}{4}$ is '$1\frac{1}{2}$ halves'

How many times $\frac{1}{2}$ go into $\frac{3}{4}$?



c $\frac{7}{3} \div \frac{5}{6} = \frac{7}{\cancel{3}^{1}} \times \frac{\cancel{6}^{2}}{5}$

$= \frac{14}{5}$

$= 2\frac{4}{5}$

d $\frac{5}{7} \div 5 = \frac{\cancel{5}^{1}}{7} \times \frac{1}{\cancel{5}^{1}}$

$= \frac{1}{7}$

Dividing by 5 is the same as multiplying by $\frac{1}{5}$.

Dividing mixed numerals

To divide mixed numerals, convert them to improper fractions first.

Example 16

Simplify $7\frac{1}{2} \div 5\frac{5}{6}$.

Solution

$7\frac{1}{2} \div 5\frac{5}{6} = \frac{15}{2} \div \frac{35}{6}$

$= \frac{\cancel{15}^{3}}{\cancel{2}^{1}} \times \frac{\cancel{6}^{3}}{\cancel{35}^{7}}$

$= \frac{9}{7}$

$= 1\frac{2}{7}$

EXERCISE 4.07 ANSWERS ON P. 573

Dividing fractions U F PS R C

1 Find the reciprocal of each fraction.

a $\frac{2}{3}$ **b** $\frac{1}{5}$ **c** $\frac{5}{8}$ **d** $\frac{1}{10}$

e 7 **f** $2\frac{1}{2}$ **g** $1\frac{3}{4}$ **h** 9

2 Evaluate each quotient.

a $4 \div \frac{1}{3}$ **b** $5 \div \frac{1}{2}$ **c** $7 \div \frac{1}{3}$ **d** $3 \div \frac{1}{5}$

e $7 \div \frac{1}{6}$ **f** $8 \div \frac{1}{10}$ **g** $4 \div \frac{1}{4}$ **h** $1 \div \frac{1}{8}$

EXAMPLE 15

3 Evaluate $\frac{3}{8} \div \frac{1}{2}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{1}{12}$ **B** $1\frac{1}{4}$ **C** $\frac{3}{4}$ **D** $\frac{3}{16}$

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4 Water is dripping from a tank at the rate of $\frac{5}{8}$ of a litre per hour. How long will it take for a 24-litre tank to empty? PS R

5 Evaluate each quotient.

a $\frac{1}{4} \div \frac{2}{3}$ **b** $\frac{8}{7} \div \frac{2}{7}$ **c** $\frac{3}{5} \div \frac{2}{5}$ **d** $\frac{5}{8} \div \frac{2}{5}$

e $\frac{7}{10} \div \frac{5}{6}$ **f** $\frac{3}{5} \div \frac{5}{8}$ **g** $\frac{11}{20} \div \frac{9}{10}$ **h** $\frac{5}{12} \div \frac{5}{6}$

i $\frac{5}{6} \div \frac{5}{12}$ **j** $\frac{1}{4} \div \frac{1}{5}$ **k** $\frac{1}{10} \div \frac{1}{2}$ **l** $\frac{1}{2} \div \frac{2}{3}$

6 **a** What number's reciprocal is itself?

b What number does not have a reciprocal?

c What is the reciprocal of the reciprocal of $\frac{7}{10}$? R

7 What happens when you: R C

a multiply any number by its reciprocal?

b divide any number by its reciprocal?

8 A bag of rice weighing $\frac{4}{5}$ kg was divided among 6 friends. How much rice did each person get? Select **A**, **B**, **C** or **D**.

A 480 g **B** $\frac{2}{15}$ kg **C** 800 g **D** $\frac{4}{15}$ kg

9 Evaluate each quotient.

a $\frac{5}{7} \div 5$ **b** $\frac{5}{7} \div 7$ **c** $\frac{1}{4} \div 3$ **d** $\frac{5}{6} \div 2$

10 Convert each mixed numeral to an improper fraction before dividing. EXAMPLE 16

a $3\frac{1}{8} \div \frac{3}{6}$ **b** $3\frac{1}{2} \div 1\frac{1}{4}$ **c** $2\frac{1}{3} \div 3\frac{1}{7}$ **d** $2\frac{1}{2} \div 1\frac{1}{8}$

11 How many speakers were there if each speaker was allowed $3\frac{1}{4}$ minutes and the debate went for about 26 minutes?

12 Copy and complete each equation. R

a $____ \div \frac{2}{3} = \frac{1}{7}$ **b** $\frac{4}{15} \div ____ = \frac{8}{21}$

13 True or false: When a number is divided by a proper fraction, the number is increased. Give reasons for your answer. R C

Investigation

Thanh's garden

In this activity, you will be using fractions to make a decorative pattern.

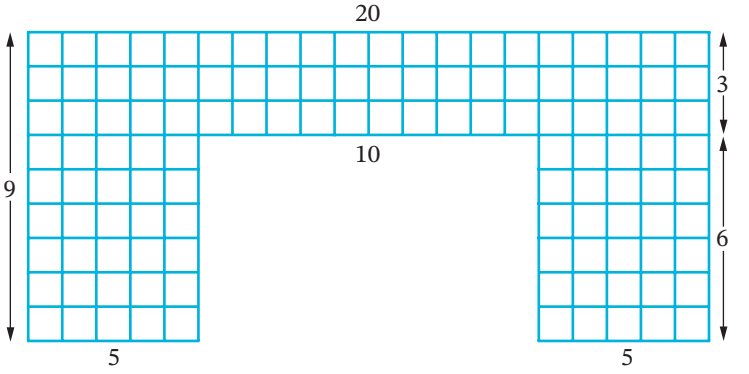
In Thanh's garden, there are places for 120 flowers. He wants to plant them in the following proportions:

$\frac{1}{3}$ golden daffodils $\frac{3}{10}$ red tulips $\frac{1}{6}$ white snowdrops $\frac{1}{5}$ blue hyacinths

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1 How many of each type of flower should Thanh plant?

2 Copy the plan below or use one provided by your teacher. Colour in squares for each flower. Make up your own pattern.



3 Make up different fractions for the same garden plot. Make a list of flowers and the fractions of their proportions and colour them on a new plan (or colour the plan first and list the fractions afterwards).

4.08 Fraction problems

EXERCISE 4.08 ANSWERS ON P. 573

Fraction problems U F PS R

1 A class investigation found that 3 out of every 25 cars passing their school was white. If 2150 cars pass the school in a day, how many would you expect to be white?

2 $\frac{3}{4}$ of every Year 7 class are boys. There are 24 students in 7 Ang, 20 in 7 Binns, 28 in 7 Carrozza and 16 in 7 Drake.

a How many boys are there in Year 7?

b How many girls are there in 7 Carrozza?

3 Anthony ate $\frac{2}{5}$ of a pizza and Lisa ate $\frac{1}{3}$.

a Who ate more?

b How much did they eat together?

c How much is left?

4 Mrs Rizzo's patio measures 5 m × 3 m. How many paving stones does it contain if there are $8\frac{1}{2}$ stones per m^2? PS

5 A bag holds $\frac{5}{8}$ of a kilogram of oranges. How many bags can $12\frac{1}{2}$ kg of oranges be packed into?

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6 When Paige walks quickly to school, she takes half an hour. When she walks slowly, she takes an extra $\frac{1}{3}$ of an hour. PS

a What fraction of an hour does Paige take walking at her slow pace?

b How many minutes is this?

7 Jeff bought 6 books and put them in his bag. His orange juice leaked and ruined $\frac{2}{3}$ of them. How many books were not ruined?

8 Mrs Chan is planning a presentation. Each slide is to be shown for $2\frac{1}{4}$ seconds. How many slides can she show in 90 seconds?

9 Each day, Mr Lanouri is at work for $7\frac{1}{2}$ hours, less $\frac{3}{4}$ of an hour for lunch. How long does he work in one day?

10 The school hall has 840 seats. If it is two-thirds full, how many seats are vacant?

11 4 authors share royalties of $\frac{1}{8}$ of the price of a textbook. Darren receives $\frac{4}{15}$ of the author share, so what fraction of the price of each book does he receive?

12 Ben was told, 'When mowing, never cut more than $\frac{1}{3}$ off the grass height'. If the lawn is $12\frac{1}{2}$ cm high, what is the minimum height to which Ben should cut the grass? Select **A**, **B**, **C** or **D**.

A $4\frac{1}{6}$ cm **B** $12\frac{1}{6}$ cm **C** $7\frac{1}{2}$ cm **D** $8\frac{1}{3}$ cm

13 200 sausages were required for a barbecue. One-tenth of them were donated by a parent, 25 had been left in the freezer from a previous occasion, and a local butcher gave one-quarter of the total free. The rest had to be bought at wholesale price. What fraction had to be bought? PS R

14 Kylie finds $\frac{1}{3}$ of her birthday cake in the fridge. If she eats $\frac{2}{3}$ of this amount, what fraction of the whole cake will be left? R

15 $\frac{1}{6}$ of a class were a day late with their assignments, while the rest handed them in on time. 20 students handed their assignments in on time. R

a How many students were in the class?

b How many students were late in handing in their assignments?

Fractions and the calculator 4.09

Scientific calculators have a fraction key for entering fractions: $\square$ or $a^{b/c}$

Casio calculators have 2 ways of entering fractions: MATH mode or LINE mode. MATH mode allows you to enter the numerator and denominator into 2 blank spaces on the calculator's screen, while LINE mode makes the fraction key act like a vinculum (fraction bar). Press SHIFT MODE to change between MATH and LINE modes. To ensure that mixed numeral answers are not converted to improper fractions, press SHIFT MODE and the down arrow to choose 'ab/c'.

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Example 17

Question	Calculator steps		Answer
	Casio LINE mode	Sharp	
Simply $\frac{16}{20}$	16 ▭ 20 =	16 $a^{b/c}$ 20 =	$\frac{4}{5}$
Change $1\frac{3}{4}$ to an improper fraction	1 ▭ 3 ▭ 4 = SHIFT S⇔D	1 $a^{b/c}$ 3 $a^{b/c}$ 4 = 2nd F $a^{b/c}$	$\frac{7}{4}$
Change $\frac{19}{4}$ to a mixed numeral	19 ▭ 4 =	19 $a^{b/c}$ 4 =	$4\frac{3}{4}$

Press SHIFT S⇔D or 2nd F $a^{b/c}$ to convert an answer between mixed numeral and improper fraction forms.

For Casio MATH mode, press the ▭ key first and use arrow keys to move the cursor to the 2 blank spaces to enter the values. Then press =.

To enter a mixed numeral, press SHIFT ▭ and move the cursor to the 3 blank spaces to enter the values. Then press =.

Example 18

Question	Calculator steps		Answer
	Casio LINE mode	Sharp	
$\frac{1}{2}+\frac{1}{3}$	1 ▭ 2 + 1 ▭ 3 =	1 $a^{b/c}$ 2 + 1 $a^{b/c}$ 3 =	$\frac{5}{6}$
$\frac{5}{7}-\frac{2}{3}$	5 ▭ 7 − 2 ▭ 3 =	5 $a^{b/c}$ 7 − 2 $a^{b/c}$ 3 =	$\frac{1}{21}$
$\frac{3}{5}\times\frac{2}{7}$	3 ▭ 5 × 2 ▭ 7 =	3 $a^{b/c}$ 5 × 2 $a^{b/c}$ 7 =	$\frac{6}{35}$
$\frac{4}{5}\div\frac{2}{3}$	4 ▭ 5 ÷ 2 ▭ 3 =	4 $a^{b/c}$ 5 ÷ 2 $a^{b/c}$ 3 =	$1\frac{1}{5}$

EXERCISE 4.09 ANSWERS ON P. 573

Fractions and the calculator U F PS R

EXAMPLE 17

1 Simplify each fraction.

a $\frac{14}{21}$ **b** $\frac{65}{130}$ **c** $\frac{27}{81}$

d $\frac{52}{65}$ **e** $\frac{168}{196}$ **f** $\frac{260}{325}$

2 Convert each mixed numeral to an improper fraction.

a $2\frac{3}{4}$ **b** $5\frac{1}{3}$ **c** $4\frac{2}{7}$

d $6\frac{3}{5}$ **e** $10\frac{1}{3}$ **f** $8\frac{5}{6}$

3 Convert each improper fraction to a mixed numeral.

a $\frac{18}{5}$ **b** $\frac{23}{7}$ **c** $\frac{45}{8}$

d $\frac{79}{11}$ **e** $\frac{123}{10}$ **f** $\frac{458}{24}$

4 Evaluate each expression.

a $\frac{3}{5} - \frac{1}{2}$ **b** $\frac{4}{7} + \frac{1}{2}$ **c** $\frac{3}{5} + \frac{2}{3}$ **d** $\frac{1}{4} \times \frac{2}{7}$

e $\frac{5}{7} \div \frac{10}{21}$ **f** $\frac{5}{8} + \frac{5}{6}$ **g** $\frac{9}{15} - \frac{1}{3}$ **h** $\frac{7}{8} \times \frac{3}{4}$

i $\frac{7}{8} \div \frac{3}{4}$ **j** $14 \div 2\frac{1}{3}$ **k** $2\frac{1}{2} + 1\frac{1}{3}$ **l** $3\frac{1}{4} - \frac{5}{8}$

m $2\frac{3}{4} \times 1\frac{1}{2}$ **n** $5\frac{2}{3} \div \frac{3}{8}$ **o** $\frac{4}{5} \div 8$ **p** $1\frac{1}{2} \times 2\frac{3}{10}$

5 Find:

a $\frac{2}{5}$ of 170 **b** $\frac{4}{7}$ of 98 **c** $\frac{1}{3}$ of 75

d $\frac{3}{8}$ of \$1360 **e** $\frac{2}{5}$ of \$4200 **f** $\frac{4}{9}$ of \$198

6 If a farm of 78 hectares is subdivided into lots of $4\frac{1}{2}$ hectares, how many lots are there?

7 Nikki has a part-time job mowing lawns. This table shows the number of hours she worked in the last month. PS

Customer	Hours
Green	$2\frac{1}{2}$
Cook	$1\frac{3}{4}$
Verbeek	$3\frac{1}{2}$
Thompson	$2\frac{3}{4}$
Burns	$2\frac{1}{4}$



Shutterstock.com/Aigars Reinholds

a Find the total number of hours Nikki worked.

b If Nikki charges \$24 per hour, how much did she earn in total?

8 Radio station FM 3.14 plays $32\frac{1}{2}$ minutes of music every hour. The average song played on the station lasts $3\frac{3}{4}$ minutes. About how many songs can be played in 24 hours? PS

9 An echidna has a mass of $1\frac{1}{2}$ kg. A field mouse has a mass of $\frac{1}{20}$ that of the echidna, but 5000 ants are needed to balance the mass of one field mouse. Find the mass of one ant. PS R

EXAMPLE 18

4.09

Foundation Standard Complex

Puzzle
Fraction arithmagons

Investigation

Fraction puzzles

Find strategies to solve each problem. Write down the thinking you used to work it out.

1 How many jelly beans?

Samantha has a bag of jelly beans to share. Ali takes $\frac{1}{4}$ of them. Duyen takes $\frac{1}{2}$ of what is left. Sharelle takes $\frac{2}{3}$ of the remainder. Samantha has 7 jelly beans left. How many jelly beans were in the bag to start with?

2 Who won the race?

- Jane finished half a minute behind Bill.
- Bill took less than half the time that Joel took.
- Joel's time was 5 minutes 23 seconds.
- Dimitra took about $3\frac{1}{4}$ minutes.
- Esther finished about $1\frac{1}{2}$ minutes after Bill.
- David finished three-quarters of a minute in front of Joel.

What is the finishing order of the six competitors?

3 How old is Zoe?

When asked her age, Zoe gave this reply:

If to my age there added be

One-half, one-third (of it) and 3 times 3,

Six score and 10 the sum you would see.

Now pray tell me what age I be.

Hint: First find how much a 'score' is.

4.10 Percentages, fractions and decimals

Worksheets
Percentages wall
Percentage shapes
Fractions, decimals and percentages

Percentages and decimals are special types of fractions. A **percentage** is a fraction whose denominator is always 100, for example, $78\% = \frac{78}{100}$. The symbol % is an abbreviation of '/100' and 'per cent', which comes from the Latin *per centum*, meaning 'by the hundred'.

A **decimal** is a fraction whose denominator is a **power of 10**, such as 10, 100, 1000; for example, $0.015 = \frac{15}{1000}$. Because fractions, percentages and decimals can be written in the form $\frac{a}{b}$, where a and b are integers but $b \neq 0$, they can all be classified as being **rational numbers**.

- $50\% = \frac{1}{2}$ because $50\% = \frac{50}{100} = \frac{1}{2}$
- $50\% = 0.5$ because $50\% = \frac{50}{100} = 50 \div 100 = 0.5$

Converting percentages to fractions and decimals

- **To convert a percentage to a fraction**, write the percentage with a denominator of 100 and simplify if needed
- **To convert a percentage into a decimal**, divide it by 100

Video
Converting percentages to fractions and decimals

Puzzles
Converting percentages to fractions
Converting fractions to percentages
Converting decimals to percentages

Presentation
Percentages, fractions and decimals

4.10

Example 19

Convert each percentage to a fraction in simplest form.

a 3% **b** 45% **c** $5\frac{1}{2}\%$

Solution

a $3\% = \frac{3}{100}$

b $45\% = \frac{45}{100}$
$= \frac{9}{20}$

c $5\frac{1}{2}\% = \frac{5\frac{1}{2}}{100}$
$= \frac{5\frac{1}{2} \times 2}{100 \times 2}$
$= \frac{11}{200}$

Example 20

Convert each percentage to a decimal.

a 39% **b** 2% **c** 16.9%

Solution

a $39\% = 39 \div 100$
$= 0.39$

b $2\% = 2 \div 100$
$= 0.02$

c $16.9\% = 16.9 \div 100$
$= 0.169$

To divide a number by 100, move the decimal point 2 places to the left

- $\frac{1}{4} = 25\%$ because $\frac{1}{4} \times 100\% = 25\%$
- $0.2 = 20\%$ because $0.2 \times 100\% = 20\%$

Converting fractions and decimals to percentages

To convert a fraction or a decimal to a percentage, multiply it by 100%.

Video
Converting fractions and decimals to percentages

Example 21

Convert each fraction to a percentage.

a $\frac{3}{4}$ **b** $\frac{1}{8}$ **c** $\frac{34}{40}$

Solution

a $\frac{3}{4} = \frac{3}{4} \times 100\%$
$= 75\%$

b $\frac{1}{8} = \frac{1}{8} \times 100\%$
$= 12\frac{1}{2}\%$

c $\frac{34}{40} = \frac{34}{40} \times 100\%$
$= 85\%$

Video
Converting fractions and decimals to percentages

Example 22

Convert each decimal to a percentage.

a 0.23 **b** 0.147 **c** 0.6

Solution

a $0.23 = 0.23 \times 100\%$
$= 23\%$

b $0.147 = 0.147 \times 100\%$
$= 14.7\%$

c $0.6 = 0.6 \times 100\%$
$= 60\%$

To multiply a number by 100, move the decimal point 2 places to the right.

EXERCISE 4.10 ANSWERS ON P. 573

Percentages, fractions and decimals U F C

EXAMPLE 19

1 Convert each percentage to a fraction in simplest form. C

a 5% **b** 14% **c** 8% **d** 90%

e 7% **f** $3\frac{1}{2}\%$ **g** $10\frac{1}{3}\%$ **h** $5\frac{3}{4}\%$

EXAMPLE 20

2 Convert each percentage to a decimal. C

a 28% **b** 6% **c** 16.1% **d** 21.3%

e 2% **f** 20% **g** 78.07% **h** $26\frac{1}{2}\%$

EXAMPLE 21

3 Convert each fraction to a percentage. C

a $\frac{9}{100}$ **b** $\frac{17}{100}$ **c** $\frac{5}{8}$ **d** $\frac{27}{30}$

e $\frac{11}{50}$ **f** $\frac{3}{10}$ **g** $\frac{1}{5}$ **h** $\frac{22}{40}$

i $\frac{3}{4}$ **j** $\frac{12}{25}$ **k** $\frac{17}{40}$ **l** $\frac{41}{50}$

m $\frac{1}{16}$ **n** $\frac{1}{3}$ **o** $1\frac{1}{2}$ **p** $\frac{2}{9}$

EXAMPLE 22

4 Convert each decimal to a percentage. C

a 0.24 **b** 0.5 **c** 0.86 **d** 0.571

e 0.06 **f** 0.345 **g** 2.4 **h** 0.007

i 1.3 **j** 0.001 **k** 0.315 **l** 0.026

5 Cameron sat for a Geography test and scored 35 out of 40. What is his score as a percentage? Select the correct answer **A**, **B**, **C** or **D**. C

A 87.5% **B** 80% **C** 35% **D** 14%

6 Which one of the following numbers is the smallest? Select **A**, **B**, **C** or **D**.

A $\frac{4}{5}$ **B** 0.885 **C** $\frac{7}{8}$ **D** 88%

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Percentage of a quantity 4.11

Percentage of a quantity

To find a percentage of a quantity, calculate:

$\frac{\text{percentage}}{100} \times \text{quantity}$

or percentage ÷ 100 × quantity

Also see **Mental skills 4B Percentage of a quantity** next page to learn how to mentally calculate with percentages.

Example 23

Find:

a 12% of 650 mL **b** 4.5% of $2400

Solution

a $12\% \text{ of } 650 \text{ mL} = \frac{12}{100} \times 650 \text{ mL}$ or $12 \div 100 \times 650$ mL

$= 78$ mL

Estimate: $12\% \text{ of } 650 \approx 10\% \times 650 = 65$

Note: If you know that 12% = 0.12, then you can also use $0.12 \times 650 \text{ mL} = 78 \text{ mL}$.

b $4.5\% \text{ of } \$2400 = \frac{4.5}{100} \times \$2400$ or $4.5 \div 100 \times \$2400$

$= \$108$

Estimate: $4.5\% \text{ of } \$2400 \approx 5\% \times \$2400 = \$120$

Note: If you know that 4.5% = 0.045, then you can also use $0.045 \times \$2400 = \$108$.

EXERCISE 4.11 ANSWERS ON P. 573

Percentage of a quantity U F

1 Find:

a	10% of 60	**b**	85% of $240	**c**	5% of 150
d	24% of 350 mm	**e**	20% of 740 kg	**f**	12% of 60 g
g	33% of $500	**h**	6% of 180	**i**	2% of $4800
j	80% of 280 mL	**k**	20% of $6.50	**l**	50% of 810 m
m	7% of $220	**n**	3% of 90 hectares	**o**	25% of 60 minutes

2 Find 5% of 74 tonnes. Select the correct answer **A**, **B**, **C** or **D**.

A 370 tonnes **B** 3.7 tonnes **C** 37 tonnes **D** 3700 tonnes

Worksheet Percentages of an amount

Puzzles Percentages

Percentages cross number

Percentages of an amount

4.11

EXAMPLE 23

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3 Erika achieved 60% in her last mathematics test. If the test was out of 70, how many marks did she receive?

4 In a school of 650 students, 48% are girls. How many girls are there in the school?

5 Farmer Chen owns 775 animals. Of these, 16% are chickens.

a How many chickens does farmer Chen have?

b How many other animals does he have?

6 Jaden planted 168 seedlings in his garden. If 12% died, how many plants survived?

7 In a city of 3 million people, 1% of the population are doctors. How many doctors are there? Select **A**, **B**, **C** or **D**.

A 3 **B** 3000 **C** 300 **D** 30 000

8 Copy and complete:

a $20\% \times _____ = \$30$ **b** $35\% \times _____ = \$56$

Quiz
Mental skills 4B

Mental skills 4B: Maths without calculators ANSWERS ON P. 573

Percentage of a quantity

Learn these commonly-used fractions and their percentage equivalents.

Fraction	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{20}$	$\frac{2}{5}$
Percentage	50%	25%	12.5%	75%	20%	10%	5%	40%

Now we will use them to find a percentage of a quantity.

1 Study each example.

a $20\% \times 45 = \frac{1}{5} \times 45$
$= 9$

b $50\% \times 28 = \frac{1}{2} \times 28$
$= 14$

c $12.5\% \times 48 = \frac{1}{8} \times 48$
$= 6$

d $5\% \times 120 = \frac{1}{20} \times 120$
$= 6$

e $75\% \times 36 = \frac{3}{4} \times 36$
$= \left(\frac{1}{4} \times 36\right) \times 3$
$= 9 \times 3$
$= 27$

f $40\% \times 35 = \frac{2}{5} \times 35$
$= \left(\frac{1}{5} \times 35\right) \times 2$
$= 7 \times 2$
$= 14$

2 Now evaluate each product.

a $75\% \times 20$ **b** $20\% \times 30$ **c** $12.5\% \times 24$ **d** $5\% \times 40$

e $10\% \times 90$ **f** $50\% \times 38$ **g** $40\% \times 40$ **h** $12.5\% \times 640$

i $25\% \times 64$ **j** $20\% \times 35$ **k** $10\% \times 170$ **l** $75\% \times 72$

m $5\% \times 600$ **n** $25\% \times 28$ **o** $50\% \times 18$ **p** $40\% \times 45$

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Did you know?

What fraction of Australia is desert?

Large parts of Australia are classified as desert, approximately $\frac{7}{20}$ or 35%.

However, $\frac{7}{10}$ or 70% of Australia is arid (dry), receiving less than 500 mm annual rainfall.

Only $\frac{3}{100}$ or 3% of the population live in these dry regions of Australia.

Find out the names of the 5 largest deserts in Australia.

Which states are they in?



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Expressing quantities as fractions and percentages

4.12

Natalie scored 34 out of 40 for an English speech. What is this mark as a percentage?

To express one quantity as a percentage of another, we need to write it as a fraction first and then convert it to a percentage by multiplying by 100%.

Worksheets
Calculating percentages
Money and percentages review

Puzzle
Calculating percentages

Expressing quantities as fractions of percentages

To write an amount as a fraction of a whole amount:

- write the amount in the numerator of the fraction
- write the whole amount (total) in the denominator
- Fraction $= \frac{\text{amount}}{\text{whole amount}}$

To write an amount as a percentage of a whole amount:

- write the amount in the numerator of a fraction
- write the whole amount (total) in the denominator
- multiply by 100%
- Percentage $= \frac{\text{amount}}{\text{whole amount}} \times 100\%$

Example 24

A school has 800 students. If 156 of them are in Year 7, find:

a the fraction of students that are in Year 7

b the percentage of students that are in Year 7.

Solution

a $\text{Fraction} = \frac{156}{800} = \frac{39}{200}$

b $\text{Percentage} = \frac{156}{800} \times 100\%$
$= 19.5\%$

So 156 is 19.5% of 800



Video
Expressing quantities as fractions and percentages

Example 25

Express:

a 34 out of 40 as a percentage

b 45 minutes as a fraction of 1 hour

c 9 hours as a percentage of 1 day

Solution

a $\text{Percentage} = \frac{34}{40} \times 100\%$
$= 85\%$

So 34 is 85% of 40.

b $\text{Fraction} = \frac{45\,\text{min}}{1\,\text{h}}$
$= \frac{45\,\text{min}}{60\,\text{min}}$
$= \frac{3}{4}$

1 hour = 60 minutes
So 45 min is $\frac{3}{4}$ of 1 hour.

c $\text{Percentage} = \frac{9\,\text{h}}{1\,\text{day}} \times 100\%$
$= \frac{9\,\text{h}}{24\,\text{h}} \times 100\%$
$= 37.5\%$

1 day = 24 hours
So 9 hours is 37.5% of 1 day.

EXERCISE 4.12 ANSWERS ON P. 574

Expressing quantities as fractions and percentages U F



1 In a road rules test, Billal answered 20 questions correctly out of 25.

a What is this score as a simplified fraction?

b If the pass mark is 92%, did Billal pass?

2 In one season a soccer team scored 20 goals. If Kirstin scored 4 of them, what is this as a percentage of the team score? Select the correct answer **A**, **B**, **C** or **D**.

A 4% **B** 5% **C** 20% **D** 40%



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3 In a class of 30 students, 12 catch the train to school, 6 walk to school and the rest are driven by their parents.

- **a** What fraction catch the train?
- **b** What percentage walk to school?
- **c** How many students are driven to school? What percentage is this?

4 Inga received the following marks in her semester exams.

Maths: 74 out of 100 | English: 28 out of 40 | Science: 50 out of 60

Geography: 20 out of 32 | Health: 13 out of 20 | Technology: 69 out of 75

- **a** Convert each test mark to a percentage.
- **b** In which subject did Inga achieve the best result?

5 Express each quantity as a percentage.

EXAMPLE 25

a	24 min of 1 h	**b**	250 mL of 1 L	**c**	700 m of 1 km
d	50c of $2	**e**	50 g of 1 kg	**f**	9 mm of 30 cm
g	6 h of 1 day	**h**	$1.75 of $7	**i**	84 h of 5 days
j	1800 g of 4 kg	**k**	1500 m of 10 km	**l**	400 mL of 2 L

Technology

Calculating test marks as percentages

Spreadsheets can easily be used to convert test marks into percentages.
Copy the data shown below into a spreadsheet.

	A	B	C	D	E	F	G	H	I	J
1	Student	Mathematics		Geography		Science		Art		
2		45	%	60	%	40	%	75	%	
3	Bond, Jeremy	33	=B3/45	45		31		60		
4	Coutler, Louise	42		50		32		55		
5	Dunlop, Nash	27		36		26		62		
6	Jones, Pip	28		52		24		70		
7	Shepherd, David	31		57		39		68		
8	Tran, Melissa	44		51		38		54		
9	Tyler, Phoebe	40		55		40		71		
10	Young, Tahlia	37		46		29		72		
11										

1. In cell C3, enter the formula **=B3/45** to convert Jeremy Bond's Mathematics mark into a decimal. Click % in the toolbar to convert the decimal into a percentage.
2. Use **Fill Down** to copy this formula into cells C4 to C10. Then highlight cells C4 to C10 and right-click on the column. Choose **Format Cells** and **Percentage**, then set **Decimal Places** to 0. This will round each percentage mark to the nearest whole number.

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3 Enter the formula **=D3/60** into cell E3 to convert Jeremy's Geography mark into a decimal, then click %. **Fill Down** to copy this formula into cells E4 to E10. Choose **Format cells**, **Percentage**, then set **Decimal Places** to 0.

4 Complete the table by using appropriate formulas to convert the Science and Art marks into percentages, rounded to the nearest whole number.

5 Now use your spreadsheet to answer each question, writing the answer in the cell shown in [square brackets].

- **a** What was the highest Mathematics percentage mark? [Answer in cell J1]
- **b** Who scored a mark of 87%? What subject was this for? [Answer in cell J2]
- **c** What was the highest mark? Which subject had this mark? [J3]
- **d** How many students achieved over 80% in Art? [J4]
- **e** Who scored over 85% in all subjects? [J5]
- **f** What was Nash's worst subject? [J6]

6 Which subject do you think had the best marks overall? In cell A11, enter the label **Average** and in cell C11, enter the formula **=AVERAGE(C3:C10)** to calculate the average of the Mathematics percentage marks. Click % in the toolbar to convert the decimal to a percentage. Choose **Format cells** and **Percentage**, then set **Decimal Places** to 1 to round the percentage to one decimal place.

7 Select cell C11 to **Copy** the formula, then **Paste** it into cells E11, G11 and I11 separately. This will calculate the averages for the Geography, Science and Art marks.

8 Which of the 4 subjects had the highest average percentage mark? Write your answer in cell J7.



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Power plus ANSWERS ON P. 574

1 Evaluate each expression.

a $\left(\frac{2}{5}\right)^2$ **b** $\left(\frac{1}{3}\right)^3$ **c** $\left(\frac{1}{2}\right)^3$ **d** $\left(\frac{7}{10}\right)^2$ **e** $\left(\frac{3}{4}\right)^3$

f $\sqrt{\frac{4}{9}}$ **g** $\sqrt{\frac{25}{49}}$ **h** $\sqrt{\frac{81}{100}}$ **i** $\sqrt{\frac{9}{16}}$ **j** $\sqrt[3]{\frac{27}{64}}$

2 Only $\frac{2}{5}$ of an iceberg is visible above the water. If the section above the water is 30 m high, find:

a the total height of the iceberg

b the height of the section below the water.

3 Todd has $\frac{3}{4}$ of a packet of biscuits. He eats half of them and has six biscuits left.

How many were in the packet originally?

4 Rukaya sold all of her eggs. She first sold half her eggs plus half an egg to her neighbour, without having to break an egg. Then she sold half her remaining eggs plus half an egg to her uncle, again without breaking an egg. Finally she sold half her remaining eggs plus half an egg to her friend, without breaking an egg. How many eggs did Rukaya begin with?

5 Simplify each fraction.

a $\frac{1}{\left(\frac{1}{2}\right)}$ **b** $\frac{1}{\left(\frac{4}{5}\right)}$ **c** $\frac{1}{1+\frac{1}{1+\frac{1}{2}}}$

6 Convert each fraction to a percentage.

a $\frac{1}{3}$ **b** $\frac{1}{6}$ **c** $\frac{1}{7}$ **d** $\frac{1}{9}$

e $\frac{1}{11}$ **f** $\frac{1}{12}$ **g** $\frac{2}{3}$ **h** $\frac{4}{9}$

7 Find:

a 87.5% of \$300 **b** $2\frac{1}{2}$% of 80 kg **c** $33\frac{1}{3}$% of 36 L

CHAPTER 4 REVIEW

Quiz
Language of maths 4

Worksheet
Fractions and decimals review

Puzzle
Fractions and percentages crossword

Language of maths

common	decimal	denominator	equivalent
fraction	improper	mixed numeral	numerator
order	percentage	proper	reciprocal
reduce	simplify	vinculum	

1 What is the bottom part of a fraction called?

2 What is the difference between a proper and improper fraction?

3 Why do you think a 'mixed numeral' has that name?

4 What does 'simplifying' or 'reducing' a fraction mean?

5 Look up the words 'reciprocal' and 'reciprocate' in the dictionary. How does what you find relate to the mathematical meaning of each word?

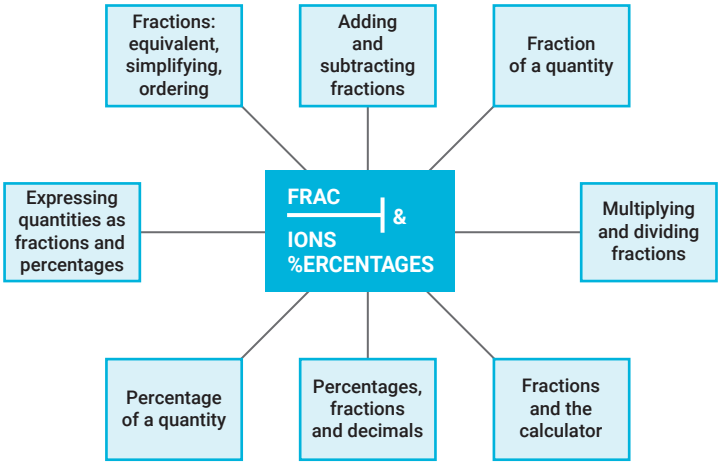
6 The 'cent' in percentage means 100. Find other words that use 'cent' and are related to 100.

Topic summary

Worksheet
Mind map: Fractions and percentages

- What have you learnt about fractions and percentages?
- Is there anything you did not understand? Ask a friend or your teacher for help.
- Where are fractions used? Give 3 examples.
- Where are percentages used? Give 3 examples.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 4

ANSWERS ON P. 574

1 Write each mixed numeral as an improper fraction. 4.01

a $1\frac{3}{4}$ **b** $2\frac{4}{9}$ **c** $5\frac{3}{5}$ **d** $3\frac{1}{7}$

2 Write each improper fraction as a mixed numeral. 4.01

a $\frac{7}{4}$ **b** $\frac{26}{3}$ **c** $\frac{23}{7}$ **d** $\frac{36}{24}$

3 Copy and complete each pair of equivalent fractions. 4.01

a $\frac{1}{3}=\frac{?}{15}$ **b** $\frac{3}{4}=\frac{12}{?}$ **c** $\frac{4}{5}=\frac{?}{20}$

d $\frac{5}{12}=\frac{15}{?}$ **e** $\frac{1}{2}=\frac{?}{50}$ **f** $\frac{3}{20}=\frac{?}{100}$

4 Simplify each fraction. 4.01

a $\frac{25}{30}$ **b** $\frac{30}{40}$ **c** $\frac{15}{20}$

d $\frac{20}{15}$ **e** $\frac{24}{48}$ **f** $\frac{24}{42}$

5 Order each set of fractions, from smallest to largest. 4.02

a $\frac{4}{5},\frac{3}{10},\frac{2}{5},\frac{7}{10}$ **b** $\frac{1}{2},\frac{1}{3},\frac{7}{12},\frac{11}{12}$

6 Evaluate each expression. 4.03

a $\frac{3}{4}+\frac{1}{4}$ **b** $\frac{2}{5}+\frac{3}{10}$ **c** $\frac{5}{10}-\frac{1}{3}$

d $\frac{3}{4}-\frac{1}{5}$ **e** $\frac{1}{2}+\frac{1}{3}$ **f** $\frac{1}{2}-\frac{3}{8}$

7 Evaluate each expression. 4.04

a $10-\frac{1}{3}$ **b** $2\frac{5}{6}+1\frac{1}{2}$ **c** $3\frac{2}{3}-2\frac{5}{6}$

d $3+2\frac{1}{2}-1\frac{1}{5}$ **e** $6-4\frac{2}{3}$ **f** $2\frac{5}{6}+4\frac{1}{2}-3\frac{2}{3}$

8 Evaluate each expression. (The letters are not in order.) 4.05

A $\frac{3}{7}$ of 21 **B** $\frac{2}{5}$ of 20 **R** $\frac{4}{9}$ of 27

O $\frac{1}{4}$ of 28 **T** $\frac{2}{3}$ of 30 **H** $\frac{1}{10}$ of 35

Order the 6 answers, beginning with the smallest. Replace each answer by its letter. What word do you spell?

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4.05 **9** Evaluate each expression.

a $\frac{2}{3}$ of 1 day (in hours) **b** $\frac{4}{5}$ of 2 km (in metres) **c** $\frac{3}{4}$ of 3 hours (in min)

d $\frac{7}{10}$ of 15 cm (in mm) **e** $\frac{1}{6}$ of 5 minutes (in seconds) **f** $\frac{5}{8}$ of 2 m (in cm)

4.06 **10** Evaluate each product.

a $\frac{2}{3} \times \frac{1}{5}$ **b** $\frac{3}{5} \times 10$ **c** $\frac{4}{7} \times \frac{5}{8}$

d $\frac{1}{3} \times \frac{1}{7}$ **e** $1\frac{1}{2} \times \frac{1}{2}$ **f** $\frac{1}{5} \times 2\frac{1}{2}$

g $1\frac{1}{2} \times 1\frac{1}{2}$ **h** $2\frac{2}{7} \times 3\frac{1}{2}$ **i** $3\frac{1}{5} \times \frac{3}{8}$

4.07 **11** Evaluate each quotient.

a $10 \div \frac{1}{5}$ **b** $\frac{1}{2} \div 3$ **c** $\frac{1}{2} \div \frac{1}{3}$

d $\frac{3}{4} \div \frac{3}{8}$ **e** $\frac{3}{5} \div \frac{1}{10}$ **f** $\frac{4}{7} \div \frac{5}{8}$

g $1\frac{1}{2} \div \frac{1}{2}$ **h** $2\frac{1}{4} \div 1\frac{1}{8}$ **i** $1\frac{7}{8} \div 1\frac{1}{4}$

4.08 **12** The recipe for a fruit punch is 2 litres of ginger ale, $1\frac{1}{2}$ litres of orange juice, $2\frac{3}{4}$ litres of lemonade and $\frac{1}{3}$ of a litre of raspberry juice. How many litres of punch does this recipe make?

4.08 **13** A farmer owns 24 goats. $\frac{1}{4}$ of the goats are black, $\frac{2}{3}$ are white, and the rest are divided equally between brown and spotted. How many of the goats are spotted?

4.09 **14** Evaluate each expression.

a $\frac{13}{18} - \frac{3}{8}$ **b** $\frac{5}{6} + \frac{4}{5}$ **c** $3\frac{1}{4} - 2\frac{3}{4}$

d $\frac{9}{16} \times \frac{2}{45}$ **e** $\frac{17}{20} \div \frac{3}{5}$ **f** $2\frac{1}{3} \times 4\frac{2}{3}$

4.10 **15** Convert each percentage to:

i a fraction **ii** a decimal

a 13% **b** 8% **c** $7\frac{1}{2}\%$

4.10 **16** Convert each to a percentage.

a $\frac{7}{10}$ **b** $\frac{23}{25}$ **c** $\frac{18}{80}$

d 0.8 **e** 0.361 **f** 0.04

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17 Copy and complete this table. 4.10

Fraction	Decimal	Percentage
$\frac{3}{5}$		
	0.8	
		82%

18 Find: 4.11

a 40% of 90 **b** 12% of $600 **c** 5% of 120

d 11% of $210 **e** 34% of 150 kg **f** 65% of 360 m

19 In one hour of TV, 22% of the time was devoted to advertising. How many minutes were devoted to advertising? 4.11

20 The netball team scored 26 goals. Jessica scored 8 goals. What fraction of her team's score did Jessica make? 4.12

21 What percentage is 300 m of 1.5 km? 4.12

22 Anand scored 51 out of 60 for an Art project. What is his mark: 4.12

a as a simple fraction? **b** as a percentage?

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